Tight Constraint on Local Dark Matter Density from a Joint Analysis of Local Kinematics and Rotation Curves

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ABSTRACT

Dark Matter (DM) particles, by their definition, are very weakly interacting and cannot be seen. They make their presence felt primarily through gravity. In particular, in the Milky Way (MW), we estimate the distribution of DM (and its velocities) by using visible matter tracers (like luminous compact old stars, extended mass like globular clusters, pulsating variable stars etc.) to construct rotation curves and mass of Halo. These rotation curves are also dependent on absolute normalisation given by the Local Standard of Rest (LSR) whose values are again dependent on visible matter tracers. Moreover, the gravitational potential in which the tracers move depend on both the distributions of dark and visible matter which are difficult to disentangle. We show that choice of the LSR, the final rotation curve made out of binning and averaging of individual tracer velocities as well as any prior on the visible matter distribution crucially influences the estimate of the DM density. This results in the wide range of values for the local DM density (as diverse as $0.1 - 0.9\mathrm{GeVcm}^{-3}$, but with small error bars) quoted in the literature. Precise knowledge of the distribution of 'visible' matter by the ongoing GAIA DR3 survey and the futuristic THEIA mission is, therefore, of paramount importance in our understanding of DM. We have also analyzed Mass distribution of our galaxy following NFW profile. This is especially important when using the astrophysical inferences about DM as inputs in particle DM experiments.

I. INTRODUCTION

One of the most popular candidates for dark matter particles is the hypothetical weakly interacting massive particle (WIMP). The determination of the density of these particles in the halo of the Milky Way galaxy in general and the solar neighborhood in particular is crucial for many direct detection experiments which measure the rate of nuclear recoil events when the WIMPs scatter off the nuclei of the detector materials, since this rate has a direct dependence on the local density.

One approach of determining the densities is to use the rotation curve (RC) data to find the likelihood of the parameters characterizing the density distributions of the various mass components of the galaxy. In general, the visible matter (VM) parameters are fixed (from observational data) and the dark matter (DM) parameters are minimised. A full likelihood analysis (DM and VM) was first done [1] by taking the Navarro-Frenk-White (NFW) profile [2] for the DM halo, a spheroidal bulge and an axisymmetric disk [3-7]. The halo and disk profiles are normalized to the respective densities at the solar neighborhood. The NFW profile is given by $\rho_{\rm DM}(r) = \rho_{\rm DM,\odot}\left(\frac{R_\odot}{r}\right)\left(\frac{r_s+R_\odot}{r_s+r}\right)^2$ where $\rho_{\rm DM,\odot}$ is the local DM density, r_s is the scale radius of the halo and $R_\odot$ is the solar distance from the galactic center. The bulge and disk density profiles are given respectively by $\rho_b(r) = \rho_{b0}\left(1+\frac{r^2}{r_b^2}\right)^{-\frac{3}{2}}$ and (in cylindrical coordinates) $\rho_d(R,z) = \frac{\Sigma_\odot}{2z_d}\exp\left(-\frac{R-R_\odot}{Rd} - \frac{|z|}{z_d}\right)$, where ρ_{b0} is the central bulge density, $\Sigma_\odot$ is the local disk surface density, and r_b and R_d are the scale radii of the bulge and the disk respectively. The parameter z_d is the scale height of the disk.

From the densities, we can obtain the total gravitational potential $\Phi(R)$ at a given radius on the galactic plane, and thus we get the circular rotation speed at R by $v_c^2(R) = R\frac{\partial}{\partial R}\Phi(R, z=0)$

Now, a Markov Chain Monte Carlo (MCMC) analysis is done using the RC data, and the most likely values of the above parameters (along with their 1σ ranges) is determined. The χ^2 test statistic used is $\chi^2 = \sum_{i=1}^{N}\left(\frac{v_{obs,i}-v_{\rm theo,i}}{\sigma_i}\right)^2$, where $v_{obs,i}$ is the observed circular velocity value, $v_{\rm theo,i}$ is the one theoretically calculated, σ_i the error in the observed velocity value, and N is the total number of data points. The likelihood estimation of the parameters have been performed with suitable prior ranges on the parameters. The value of z_d is immaterial to our analysis, and it has been fixed [8] at 340 pc.

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II. MILKY WAY MASS MODEL & LOCAL DARK MATTER

We model the Galaxy in terms of three components:

- A spherical dark matter (DM) Halo,
- A spheroidal visible matter (VM) Bulge, and
- An axisymmetric VM disk.

We assume a DM halo of the Navarro-Frenk-White (NFW) form [2], where the density ρ_{DM} of the halo as a function of galactocentric radius r is given by

$$\rho_{\mathrm{DM}}(r) = \rho_{\mathrm{DM},\odot}\left(\frac{R_\odot}{r}\right)\left(\frac{r_s+R_\odot}{r_s+r}\right)^2 \tag{1}$$

The VM bulge profile density is given by

$$\rho_b = \rho_{b0}\left(1+\frac{r^2}{r_b^2}\right)^{\frac{-3}{2}} \tag{2}$$

while the disk density profile is given by

$$\rho_d(R,z) = \frac{\Sigma_\odot}{2z_d}\exp\left(-\frac{R-R_\odot}{R_d}-\frac{|z|}{z_d}\right) \tag{3}$$

Given a spherically symmetric density distribution $\rho(r)$, the mass enclosed inside a sphere of galactocentric radius r can be written as

$$M(r) = \int_0^\infty 4\pi r'^2\rho\left(r'\right)dr' \tag{4}$$

For the NFW profile, it is given by

$$M_{\mathrm{DM}}(r) = 4\pi\rho_{\mathrm{DM},\odot}R_0\left(r_s+R_0\right)^2\left(\log\frac{r_s+r}{r_s}-\frac{r}{r+r_s}\right) \tag{5}$$

while for the central bulge, it is given by

$$M_b(r) = 4\pi\rho_{b0}\left[\sinh^{-1}\left(\frac{r}{r_b}\right)r_b^3-\frac{rr_b^4\sqrt{1+\left(\frac{r}{r_b}\right)^2}}{r^2+r_b^2}\right] \tag{6}$$

The rotational speed v_c^2 is then given by

$$v_c^2 = \frac{GM(r)}{r} \tag{7}$$

The disk is not spherically symmetric, and the calculation of $v_c^2(r)$ is not very straightforward. The steps for this are outlined by Duric, and we show the result here,

$$v_{c,disk}^2(r) = 4\pi G\Sigma_\odot R_d\left(\frac{r}{2R_d}\right)^2\exp\left(\frac{R_\odot}{R_d}\right)\times \left[I_0\left(\frac{r}{2R_d}\right)K_0\left(\frac{r}{2R_d}\right)-I_1\left(\frac{r}{2R_d}\right)K_1\left(\frac{r}{2R_d}\right)\right] \tag{8}$$

where $I_n(x)$ and $K_n(x)$ are the Modified Bessel functions of order n of the first and second kinds respectively. We perform the analysis imposing the following priors on the VM disk -

- No prior.
- Gausssian Prior 1 : $\Sigma_\odot = 48\pm 8\mathrm{M_\odot pc^{-2}}, R_d = 2.3\pm 0.6\mathrm{kpc}$
- Gaussian Prior 2 : $\Sigma_\odot = 67\pm 8\mathrm{M_\odot pc^{-2}}, R_d = 2.3\pm 0.6\mathrm{kpc}$

III. CONNECTING TRACERS TO DM DISTRIBUTION WITH ROTATION CURVE OF GALAXY :

Bhattacharjee et al.(2014) uses several tracers like Cepheids, Planetary Nebulae, Clusters etc. to map the rotation curve of the Milky Way galaxy. Prior to the Gaia mission and the availability of accurate proper motions for several stars, the only available data was spectroscopically determined Line-of-Sight(LoS) velocities.

These LoS velocities can be used to reconstruct the rotation curve with the assumption that the tracers all orbit the Milky Way in circular orbits. One of the tracers used are Classical Cepheids, which have accurately determined distances from the Period-Luminosity (PL) relations. Data from Pont et al.(1994) which provide gamma velocities (radial velocities accounting for the pulsation of the surface) and use Period-Luminosity-Colour (PLC) relations and reddening data to determine the distance to the stars.

We have knowledge now of the LoS velocities (v_h), and the distance to the stars (r_h). Since we observe from the Sun's frame, we also need information about the Sun's distance from the Galactic Centre, and the velocity of the Sun from the frame of the Galactic Center.

The radial distance (R_0) from the Galactic Centre (GC), and the Circular Velocity (V_0) of the Sun i.e. $[R_0, V_0]$ defines the Local Standard of Rest (LSR) which follows the mean motion of material near the Sun in the Milky Way. The Sun's motion is also characterized by its Peculiar Velocity ($U_\odot, V_\odot, W_\odot$) with respect to the LSR.

Shifting from the Sun's frame to the frame of an observer in the LSR, we can write

$$v_{LSR} = v_h + U_\odot\cos b\cos l + V_\odot\cos b\sin l + W_\odot\sin b \tag{9}$$

This accounts for the Sun's peculiar motion, which is known from measuring the motion of the stars in the vicinity of the Sun. Now, we need to move from the LSR to the Galactocentric Frame, noting that the LoS velocity measured is the projection of the actual (assumed) circular velocity along the loine of sight. With knowledge of the galactic coordinates (l, b), their relation can be derived as

$$V_c(R) = \frac{R}{R_0}\left[\frac{v_{LSR}}{\sin l \cos b} + v_0\right] \tag{10}$$

where R is the distance of the source from the GC, which is given by

$$R = \sqrt{R_0^2 + r_h^2\cos^2 b - 2R_0 r_h \cos b \cos l} \tag{11}$$

To properly see the features of the Rotation Curve, we bin the data and average the samples in each bin to get a mean V_c and mean R for each bin. These values are then plotted.

The coverage of the data in the galactic plane can be visualized by a scatterplot between the X and Y coordinates.

$$\begin{aligned} x &= r_h \cos b \sin l \\ y &= R_0 - r_h \cos b \cos l \\ z &= r_h \sin b \end{aligned} \tag{12}$$

Using Jean's Equation for Halo Tracers I

Using Jean's equation (Binney & Tremaine, 2008), we can write

$$V_c^2 = -\sigma_R^2\left[\frac{d\ln n_{tr}}{d\ln r} + \frac{d\ln\sigma_R^2}{d\ln r} + 2\beta\right] \tag{13}$$

where, $\sigma_R\,(\sigma_T)$ is the velocity dispersion in the radial(transverse) direction.

$$\sigma_T^2 = \frac{\sigma_\phi^2 + \sigma_\theta^2}{2} \tag{14}$$

and β is the anisotropy defined as

$$\beta = 1 - \frac{\sigma_T^2}{\sigma_R^2} \tag{15}$$

n_{tr} is the tracer density given by

$$n_{tr} = \frac{\text{Number of tracers in a bin}}{4\pi R^2 dR}$$

To calculate the velocities in the $(\mathrm{R}, \theta, \phi)$ coordinates, we note that the velocities $(\mathrm{U}_2,\ \mathrm{V}_2,\ \mathrm{W}_2)$ of the tracer in the Galactocentric frame are along the $(\ \mathrm{X}, \mathrm{Y}, \mathrm{Z}\ )$ coordinate system defined with the Sun at $(0, \mathrm{R}_0, 0)$ (upto a change in sign). Hence we can convert between them.

The 6-D phase parameters provided by the Gaia-DR3 catalog for each star contains the parallax π, the Galactic coordinates (ℓ, b), the radial velocity v_r and the two proper motions in equatorial coordinates $\mu_\alpha \cos\delta$ and μ_δ (which can be transformed into Galactic coordinates as $\mu_\ell \cos b$ and μ_b). These six variables allow for each star to be placed in the 6D phase space: 3D spatial +3 D velocity.

The Galactocentric position in cylindrical coordinates has the following three components: the Galactocentric distance R, the Galactocentric azimuth ϕ[1], and the vertical distance Z. Analogously, the Galactocentric velocity in cylindrical coordinates has the following three components: the radial velocity V_R, the azimuthal velocity V_ϕ, and the vertical velocity V_Z.

$$\begin{aligned} r &= \frac{1\mathrm{AU}}{\pi} \\ X(r,\ell,b) &= R_\odot - r\cos\ell\cos b, \\ Y(r,\ell,b) &= r\sin\ell\cos b, \end{aligned}$$

$$\begin{aligned} Z(r,b) &= Z_\odot + r\sin b, \\ R(r,\ell,b) &= \sqrt{X^2+Y^2} \\ \phi(r,\ell,b) &= \tan^{-1}\left(\frac{Y}{X}\right) \quad \left(\text{between } \frac{\pi}{2} \text{ and } \frac{3\pi}{2} \text{ for } X<0\right), \end{aligned}$$

$$\begin{aligned} &\textit{and } \boldsymbol{V}(r,\ell,b,v_r,\mu_\ell,\mu_b) = \\ &\begin{pmatrix} V_R \\ V_\phi \\ V_z \end{pmatrix} = \begin{pmatrix} -\cos\phi & \sin\phi & 0 \\ \sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} U_* + U_\odot \\ V_* + V_{g,\odot} \\ W_* + W_\odot \end{pmatrix} \\ &\begin{pmatrix} U_* \\ V_* \\ W_* \end{pmatrix} = \begin{pmatrix} \cos\ell\cos b & -\sin\ell & -\cos\ell\sin b \\ \sin\ell\cos b & \cos\ell & -\sin\ell\sin b \\ \sin b & 0 & \cos b \end{pmatrix} \begin{pmatrix} v_r \\ v_\ell \\ v_b \end{pmatrix} \\ &v_\ell = r\mu_\ell\cos b \\ &v_b = r\mu_b \end{aligned}$$

The Galactic parameters used by the previous kinematic Gaia-DR3 analysis (G18): $R_\odot = 8.34$kpc (Reid et al. 2014), $Z_\odot = 0.027$kpc (Chen et al. 2001); $(U_\odot, V_\odot, W_\odot) = (11.1, 12.24, 7.25)$km/s (Schönrich et al. 2010); $V_{g,\odot} = V_{c,\odot} + V_\odot$, with the rotation speed $V_{c,\odot} =$ 240 km/s

We have knowledge of proper motions in RA (μ_α) and DEC (μ_δ) and their errors and covariances from Gaia DR3, i.e. the covariance matrix,

$$\Sigma_{\alpha\delta} = \begin{pmatrix} \sigma_{\mu_\alpha}^2 & \sigma_{\mu_\alpha\mu_\delta} \\ \sigma_{\mu_\alpha\mu_\delta} & \sigma_{\mu_\delta}^2 \end{pmatrix} \tag{16}$$

Transformed to galactic coordinates (Poleski, 2013), we get

$$\begin{pmatrix}\mu_l\\ \mu_b\end{pmatrix} = \frac{1}{\cos b}\begin{pmatrix} C1 & C2\\ -C2 & C1\end{pmatrix}\begin{pmatrix}\mu_\alpha\\ \mu_\delta\end{pmatrix} \tag{17}$$

and the covariance matrix transforms as

$$\Sigma_{lb} = C\Sigma_{\alpha\delta}C^T \quad (18) \qquad where$$

$$\begin{aligned} C1 &= \sin\delta_G\cos\delta - \cos\delta_G\sin\delta\cos(\alpha-\alpha_G)\\ C2 &= \cos\delta_G\sin(\alpha-\alpha_G)\\ \cos b &= \sqrt{C1^2 + C2^2}\\ C &= \frac{1}{\cos b}\begin{pmatrix} C1 & C2\\ -C2 & C1\end{pmatrix}\end{aligned}$$

Now, we find the 3-D velocities (in kms^{-1}), which requires LoS distance r_h, of the tracers as

$$\begin{aligned} V_l &= 4.74 r_h \mu_l\\ V_b &= 4.74 r_h \mu_b\end{aligned}$$

The new covariance matrix is given by

$$\begin{aligned}\Sigma_{V_{lb}} &= F\Sigma_{lb}F^T = \begin{pmatrix}\sigma^2_{V_b} & \sigma_{V_lV_b}\\ \sigma_{V_lV_b} & \sigma^2_{V_l}\end{pmatrix}\\ F &= \begin{pmatrix} 4.74\mu_l & 4.74r_h & 0\\ 4.74\mu_b & 0 & 4.74r_h\end{pmatrix}\end{aligned}$$

where

$$\begin{aligned}\sigma^2_{V_l} &= V_l^2\left(\frac{\sigma_r^2}{r_h^2} + \frac{\sigma_l^2}{\mu_l^2}\right)\\ \sigma^2_{V_b} &= V_b^2\left(\frac{\sigma_r^2}{r_h^2} + \frac{\sigma_b^2}{\mu_b^2}\right)\\ \sigma_{V_lV_b} &= V_lV_b\left(\frac{\sigma_r^2}{r_h^2} + \frac{\sigma_l b}{\mu_l\mu_b}\right)\end{aligned}$$

Now, we can rotate to UVW coordinates at the Sun's position

$$\begin{pmatrix} U1\\ V1\\ W1\end{pmatrix} = \begin{pmatrix}\cos b\cos l & -\sin b\cos l & -\sin l\\ \cos b\sin l & -\sin b\sin l & \cos l\\ \sin b & \cos b & 0\end{pmatrix}\begin{pmatrix}V_r\\ V_b\\ V_l\end{pmatrix} \tag{20}$$

and the covariance matrix is

$$\begin{aligned}\Sigma_{UVW_1} &= F_1\Sigma_{V_rbl}F_1^T\\ \Sigma_{V_rbl} &= \begin{pmatrix}\sigma^2_{V_r} & 0 & 0\\ 0 & \sigma^2_{V_b} & \sigma_{V_lV_b}\\ 0 & \sigma_{V_lV_b} & \sigma^2_{V_l}\end{pmatrix}\end{aligned}$$

and F_1 is the transformation matrix in Eq.20.

Now, we can add the Sun's velocity to get the galactocentric velocites.

$$\begin{aligned} U2 &= U1 + U_\odot\\ V2 &= V1 + V_\odot + V_0\\ W2 &= W1 + W_\odot\end{aligned}$$

One final rotation to the UVW coordinates at each tracer's position will give the final values we want.

$$\begin{aligned}\begin{pmatrix}U_s\\ V_s\\ W_s\end{pmatrix} &= \begin{pmatrix}\cos\beta & -\sin\beta & 0\\ \sin\beta & \cos\beta & 0\\ 0 & 0 & 1\end{pmatrix}\begin{pmatrix}V_r\\ V_b\\ V_l\end{pmatrix}\\ \sin\beta &= \frac{r_h\cos b\sin l}{R}\\ \cos\beta &= \frac{[R_0 - r_h\cos b]\cos l}{R}\end{aligned}$$

and the final covariance matrix, using the above transformation matrix F_2, is

$$\Sigma = F_2\Sigma_{UVW_1}F_2^T$$

LOCAL KINEMATICS

For our kinematic analysis, we use data from [18]. [18] analyze the kinematics of 412 red giants, located at distances ranging from $1.5-4.5$kpc from the Galactic midplane. These 412 stars are selected so as to exclude the thin disk and galactic halo stars and ensure that they belong to the Galactic thick disk. For each star, they calculate the (U, V, W) velocity components in the Galactic cylindrical reference frame, (U is positive towards the Galactic center, V is directed in the sense of Galactic rotation and points towards the North Galactic pole). They adopt a solar peculiar velocity of (U,V,W) = (10.0, 5.1, 7.2)kms^{-1}. In their analysis, they adopt a $R_\odot = 8.0$kpc. The choice of solar motion, however does not affect the velocity dispersions. They divide the sample into 31 bins, with bin centers ranging from 1.5 to 4.5 kpc separated by 0.1 kpc , and calculate the mean (U, V, W) velocities, the dispersion velocities $(\sigma_U, \sigma_V,\ \sigma_W)$ and the non diagonal terms of the dispersion matrix $(\sigma^2_{UW}, \sigma^2_{UV}, \sigma^2_{VW})$. The errors on the velocities are derived from propagating uncertainties from observed quantities, and the errors on the dispersion velocities are derived by analyzing artificial data through Monte Carlo simulations. For our analysis, we use the σ_W and σ^2_{UW} data (denoted by σ^2_z and σ_{rz}). This data has been previously analyzed by [18] and[19]. We follow the approach described in [18] and [19]. Our analysis is based on the vertical Jean's equation

$$\frac{1}{R\nu}\frac{\partial}{\partial R}\left(R\nu\sigma_{Rz}\right)+\frac{1}{R\nu}\frac{\partial}{\partial\phi}\left(\nu\sigma_{\phi z}\right)+\frac{1}{\nu}\frac{d}{dz}\left(\nu\sigma_z^2\right)=-\frac{d\Phi}{dz} \quad (21)$$

Here Φ is the gravitational potential and ν is the stellar number density. On assuming axial symmetry for the Galaxy, the second term on the left hand side drops out, and we obtain

$$\frac{1}{R\nu}\frac{\partial}{\partial R}\left(R\nu\sigma_{Rz}\right)+\frac{1}{\nu}\frac{d}{dz}\left(\nu\sigma_z^2\right)=-\frac{d\Phi}{dz}=F_Z(R,Z) \quad (22)$$

We then write the Poisson equation in cylindrical coordinates

$$\Sigma(R,z)=-\frac{1}{2\pi G}\left[\int_0^z dz\frac{1}{R}\frac{\partial\left(RF_R\right)}{\partial R}+F_Z(R,Z)\right] \quad (23)$$

$F_R\left(=-\frac{\partial\Phi}{\partial R}\right)$ can be linked to the circular velocity as $V_C^2=-RF_R$. As $\frac{\partial V_C}{\partial R}=0$, the integral term can be dropped from the Poisson equation. However, observations show that V_C is constant only in the galactic plane. [18] show that approximating the integral in Eq. 23 by its value in the galactic plane (Z = 0) leads to an underestimation of the dark matter density. We adopt this approximation, and integrating 22 gives

$\sigma_z^2(z)=\frac{1}{\nu(z)}\int_0^z\nu\left(z'\right)\left[F_z\left(z'\right)-\frac{1}{R\nu}\frac{\partial}{\partial R}\left(R\nu\sigma_{Rz}\right)\right]dz'+\frac{C}{\nu(z)}$
where

$$2\pi G\Sigma(z)=\frac{d\Phi}{dz}=-F_z \quad (24)$$

and C is a normalization parameter. Substituting z = 0 , we have $\sigma_z^2(0)\nu(0)=C$.

Following [18], we evaluate $\frac{1}{R\nu}\frac{\partial}{\partial R}\left(R\nu\sigma_{Rz}\right)$ (the Tilt term τ) by assuming that radial profiles of the tracer density and σ_{Rz} can be separated and are exponentials, given by

$$\nu(R,z)=\nu(z)\exp\left(-\frac{R}{h_R}\right)=\nu\left(R_\odot\right)\exp\left(-\frac{z}{h_z}\right)\exp\left(-\frac{R}{h_R}\right) \quad (25)$$

$$\sigma_{Rz}(R,z)=\sigma_{Rz}(z)\exp\left(-R/h_\sigma\right) \quad (26)$$

We assume $h_R=h_\sigma=3.8$kpc same as [18], and $h_z=0.9$kpc. Under these approximations, the tilt term reduces to

$$\tau(R,z)=\sigma_{Rz}(R,z)\left[\frac{1}{R}-\frac{1}{h_R}-\frac{1}{h_\sigma}\right] \quad (27)$$

We assume a linear profile for $\sigma_{Rz}(z)$ [18] and as we calculate the velocity dispersions at the solar radius, we have

$$\sigma_{Rz}\left(R_\odot,z\right)=A+B(z-2.5) \quad (28)$$

We fit the Moni Bidin's data with this profile and obtain best fit values of A = 1033.88 ± 64.92 and B = 537.53 ± 73.53.

Combining all this, we have

$$\sigma_z^2(z)=\frac{1}{\nu(z)}\int_0^z\nu\left(z'\right)\left[-2\pi G\Sigma\left(z'\right)-\sigma_{Rz}\left(R_\odot,z'\right)\left[\frac{1}{R_\odot}-\frac{1}{h_R}-\frac{1}{h_\sigma}\right]\right]dz'+\sigma_z^2(0) \quad (29)$$

This is the equation central to the analysis described by [18], and is identical to equation (25) of [18] in their analysis .

We split $\Sigma(z)$ as

$$\Sigma(z)=\Sigma_{DM}(z)+\Sigma_{\text{Baryon}}(z) \quad (30)$$

Following the method described by [18], we use Multinest to perform a Bayesian analysis on the [18] σ_z^2 data and estimate the parameters in our models for Σ_{DM} and Σ_{Baryon} . We model the local mass density using a single exponential disk, and a NFW dark matter halo.

LOCAL KINEMATICS & ROTATION CURVE INDIVIDUAL -VS- JOINT ANALYSIS

INDIVIDUAL CONSTRAINTS

DATASETS, BINNING AND AVERAGING

A. Bhattacharjee's Data Selection

- Two independent classes of non-disk stellar tracers-
  1. 4985 BHB stars (SDSS DR8, Xue et al.), 4781 K Giants (SDSS DR9, Xue et al.)- for galactic halo to 100 kpc
  2. Heterogeneous sample of 430 objects- 143 Globular Clusters (Harris, 2010), 118 Red halo giants (Carney et al. 2003, 2008), 108 field blue horizontal branch (Clewley et al. 2004), 38 RR Lyrae (Kinman et al. 2012), 23 dwarf spheroidals (McConnachie 2012)
  3. Cut on z and R of tracers, leaving out r < 25 kpc.
  4. Final sample: 1457 BHB, 2227 K Giants; 16 Globular Clusters, 28 FHB stars, 21 dwarf spheroidals (grouped as Hg)
- BHB- uniform bin widths, 2 kpc over 25 to 55 kpc.

- KG- bin width = 2 kpc for 25 to 55 kpc, and 4 kpc for 55 to 103 kpc, anything greater than 103 kpc is one bin.
- Hg- object-wise binning, first 6 radial bins with 8 objects each, next 2 bins with 6 objects each, and remaining 5 objects in 1 bin.

B. Eiler's Data Selection

- Stars are chosen from upper red giant branch with low surface-gravity, i.e. with a surface gravity of $0 \leq \log g \leq 2.2$, which selects stars that are more luminous than the red clump.
- These need to have existing spectral data from APOGEE and complete G-band information from Gaia, near-infrared from + 2MASS, and W1/W2 infrared bands from WISE (at 3.6 and 4.5 micrometer respectively).
- Data quality cuts are made to give 44,784 stars.
- For circular velocity curve:
  1. Stars within a wedge of 60∘ from the Galactic center towards the direction of the Sun, i.e. ±30∘
  2. $|z| \leq 0.5$ kpc or lie within 6∘ from the Galactic plane, i.e. $|z|/R \leq \tan(6\circ)$
  3. Exclude stars with a vertical velocity component of $|v_z| >100$ km s^{-1}
  4. Stars with low α-element abundances, i.e.[α/Fe]< 0.12
  5. Gives a total of 23,189 stars for analysis.

C. Mroz Data Selection

Data from Skowron et al. (2018)- 2177 Galactic Cepheids.Cross-match with Gaia DR2 which gives full information for 832 objects. Known cepheids in binary systems, objects with residual velocities at least 4σ times larger than mean, where is the dispersion of residuals. Cepheids observed from 2 to 44 times with the median number of seven visits were kept.Gives a final set of 773 Cepheids.

To determine the RC of the galaxy, one has to use the kinematical and positional information for some tracer objects moving in the gravitational potential of the galaxy. In general, one measures the line-of-sight (LOS) quantities (positional and kinematical data) and the RC is derived from that. To determine the RC for the disk region, we have to adopt a value for the the local standard of rest (LSR) which corresponds to the position (R_0) and velocity (v_{c0}) of the sun with respect to the galactic center, and make the assumption that the tracer objects follow a circular orbit arround the galactic center. From this, the positions and velocities about the galactic center can be obtained. The various tracer objects used for this include HI and HII regions (CO emissions from the latter), Cepheids, Planetary nebulae, etc. For regions extending beyond the visible disk of the galaxy, we look at tracers distributed in the halo of the Milky Way (like dwarf spheroids, globular clusters, K-giant stars, etc.). These tracers are of the non-disk kind, and their motion around the galactic center is typically unsystematic. Under the assumption that these objects are isotropically distributed in the halo, one can use the Jeans to relate their velocity v_c and their distance r from the center to their observed number densities and velocity dispersions. One disadvantage of this approach is that the velocity anisotropy is completely unknown (since observationally, only the radial velocity dispersion is seen), and can significantly affect results [9].

After the raw RC is generated as above, it is suitably binned and averaged. Different strategies for the above procedure can lead to different results. Eg., The data of Sofue 2013 [10] employs a running average method, where all the raw data points are taken, and the midpoints of the bins are $(1 + \epsilon)$ times ($\epsilon = 0.1$ or 0.3) the previous value. The weight in each bin is Gaussian, with σ being η times the bin-center value. The averaging is done only once. While calculating the mean, the observables (r or v) at higher radii are given more weight. The formula provided for calculating the error bars shows that errors at higher radii have a higher value. Bhattacharjee et.al. [9] have averaged twice, once over the individual data sets, and then over the first-averaged data, with "optimal" bin sizes in different radial ranges to capture the small-scale features of the raw RC, as well as to get the closest value to the true RC. Figures 2a and 2b show the difference in the final RC due to these strategies with the underlying data almost same [9, 10].

The inset of Fig.2 shows the 2D posterior probability contours for the DM parameters, namely $\rho_{\mathrm{DM},\odot}$ and r_s, obtained from the datasets of Sofue (2013) (henceforth referred to asS13) and Bhattacharjee et.al., both for the LSR of ($8.0\mathrm{kpc}, 200$ km s^{-1}). The larger error bars in the S13 data, especially around the outer regions of the galaxy, where the effect of the DM is strongly predominant, loosens the posterior constraints on the DM parameters. Also, in Fig.1 we see the rotation curves recreated using the obtained mean values obtained from the MCMC analysis. The blue lines (dased and solid) in either subfigure denote to the curves obtained using flat priors on the disk parameters.

Fig.2 shows the comparison between the 1σ and 2σ 2 D posterior probability contours of the DM parameters, computed for the following cases (with the mean values tabulated in Table IV):

1. Bhattacharjee et.al. $(8.0, 200)$ data (with flat priors) with (a) the full dataset, (b) data points above 3 kpc , (c) data points above 5 kpc , and (d) data points above 5 kpc and bulge neglected in the MCMC analysis.

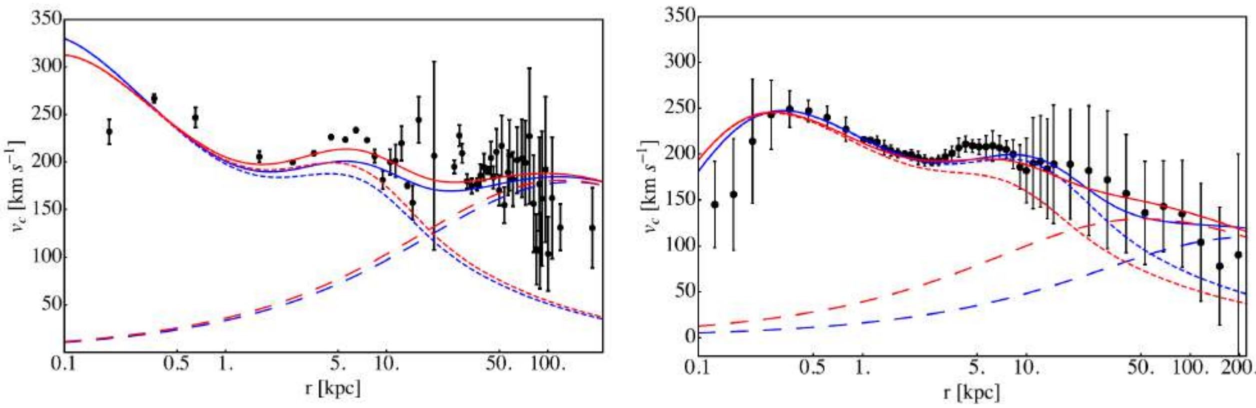

FIG. 1. Recreated RC's from the obtained values of the parameters obtained with weak (flat) and strong (Gaussian) priors on the disk parameters. The dashed (dotted) line represents the RC due to the DM (VM) components, and the color blue (red) signifies weak (strong) priors. The solid line is the combined RC.

2. Sofue. (2013) data (with flat priors) with (a) the full dataset, (b) data points above 3 kpc , (c) data points above 5 kpc .

Dataset	$\rho_{DM\odot}$ $\left(\text{GeVcm}^{-3}\right)$	r_s(kpc)
Bhatt (8.0, 200)	$0.16^{+0.01}_{-0.01}$	$64.83^{+17.03}_{-12.94}$
Bhatt (8.3, 244)	$0.05^{+0.01}_{-0.01}$	$213.15^{+164.09}_{-102.76}$
Bhatt (8.5, 220)	$0.09^{+0.02}_{-0.01}$	$74.58^{+38.24}_{-27.56}$
Sofue (2013)	$0.04^{+0.07}_{-0.02}$	$106.14^{+230.52}_{-86.18}$

TABLE I: Mean values of the DM parameters obtained with flat priors on the disk parameters.

Parameter	Flat Priors	Gaussian Priors
$\rho_{DM\odot}$ $\left(\text{GeVcm}^{-3}\right)$	$0.16^{+0.01}_{-0.01}$	$0.18^{+0.02}_{-0.01}$
r_s(kpc)	$64.83^{+17.03}_{-12.94}$	$57.21^{+140}_{-10.95}$
$\rho_{DM\odot}$ $\left(\text{GeVcm}^{-3}\right)$	$0.04^{+0.07}_{-0.02}$	$0.16^{+0.07}_{-0.08}$
r_s(kpc)	$106.29^{+230.52}_{-86.18}$	$24.16^{+41.22}_{-10.77}$

TABLE II: DM parameters obtained with and without priors on the disk parameters, respectively Bhattacahrjee et.al. (8.0), Sofue 2013.

Choice of the Local Standard of Rest

We have used three different datasets[9] corresponding to three different choices of the LSR - respectively $\left(8.0\text{kpc}, 200 \text{ km s}^{-1}\right)$, $\left(8.3\text{kpc}, 244 \text{ km s}^{-1}\right)$ and $\left(8.5\text{kpc}, 220 \text{ km s}^{-1}\right)$. We refer to these henceforth as B80, B83, and B85 respectively. The dataset of S13 [10] uses a LSR of $\left(8.0\text{kpc}, 200 \text{ km s}^{-1}\right)$.

Fig.2 shows the 1σ and 2σ2D posterior probability contours of the DM parameters ($\rho_{\text{DM},\odot}$ and r_s) for the three choices of the LSR's after marginalizing over the other (VM) parameters. The corresponding mean values are given in Table I. It is seen that the choice of the LSR has a significant effect on the posterior PDF of the parameters, with nearly exclusive zones in the $\rho_{\text{DM},\odot} - r_s$ space.

In Fig.2, we can see the comparison between the posterior probality contours obtained using different choice of LSR's for the datasets. The blue and the green contours differentiate between the choices of LSR of ($8.5\text{kpc}, 220 \text{ km s}^{-1}\big)$ and $\left(8.0\text{kpc}, 200 \text{ km s}^{-1}\right)$ for flat priors (on the disk parameters), while the respective lighter contours do so for the Gaussian priors (as specified in more details in the next section).

EFFECT OF VISIBLE MATTER PRIORS ON DM CONSTRAINTS

The addition of contraints on some of the visible matter parameters causes a significant change in the posterior probabilities of the parameters. In particular, large shifts are seen in the obtained mean values of the DM parameters. In our analysis, we have put gaussian priors [5, 11] of $\Sigma_0 = 48 \pm 8 \text{M}_\odot\text{pc}^{-2}$ and $R_d = 2.3 \pm 0.6$kpc. Table II lists the obtained mean values and 1σ ranges of the DM parameters obtained from the Bhattacharjee et.al data set $(8, 200)$, and the Sofue 2013 data sets, and Figure 1 shows the theoretically recreated RC's using these values of the parameters.We have obtained the 1σ and 2σ2D posterior probability contours of (i) the DM parameters, (ii) the $\rho_{\text{DM},\odot} - R_d$ parameters, and (iii) the $\rho_{\text{DM},\odot} - \Sigma_\odot$ parameters obtained in each case with and without the use of priors on the disk densities, for the Bhattacharjee et.al dataset for three LSR's - (a) $\left(8.3\text{kpc}, 244 \text{ km s}^{-1}\right)$, (b) $\left(8.5\text{kpc}, 220 \text{ km s}^{-1}\right)$, and (c) $\left(8.0\text{kpc}, 200 \text{ km s}^{-1}\right)$. The obtained mean values for these parameters are tabulated in Table III.

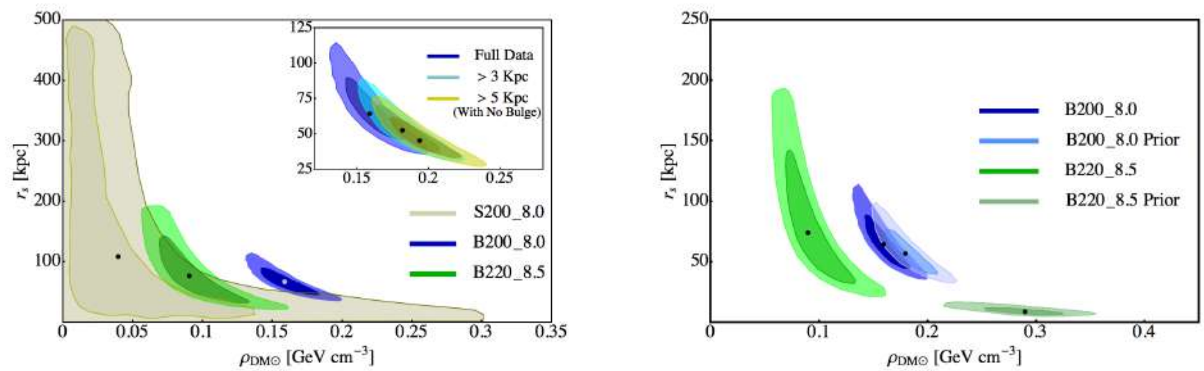

FIG. 2. Comparison of posterior probability density with different LSRs with prior i.e. flat prior (left) and with prior(right).

Dataset	$\rho_{DM\odot}$ (GeVcm^{-3})	r_s(kpc)	$\Sigma_\odot$ (M$_\odot$pc^{-2})	R_d(kpc)
Bhatt. et.al. (8.0, 200)	0.16(0.18)	64.83(57.21)	53.37(60.72)	2.75(2.72)
Bhatt. et.al. (8.3, 244)	0.05(0.37)	213.15(7.53)	213.33(119.88)	4.73(3.91)
Bhatt. et.al. (8.5, 220)	0.09(0.29)	74.58(8.81)	138.88(77.56)	4.26(3.60)

TABLE III: DM and VM parameters obtained with flat (Gaussian) priors on the disk parameters.

Dataset	$\rho_{DM\odot}$ (GeVcm^{-3})	r_s(kpc)
Bhatt 8.0 (Full)	0.16	64.83
Bhatt 8.0 (Above 3 kpc)	0.18	52.82
Bhatt 8.0 (Above 5 kpc, No Bulge)	0.19	46.5
Sofue 2013 (Full)	0.04	106.14
Sofue 2013 (Above 3 kpc)	0.06	65.76

TABLE IV: DM parameters obtained with flat priors on the disk parameters on different modified versions of the datasets.for comparison see FIG. 2

Parameter	**(8.0,200)**	**(8.3,244)**	**(8.5,220)**	**Eilers**	**Mroz**
$\rho_{\rm DM,0}$ [GeV cm^{-3}]	0.32	0.26	0.37	0.38	0.432
r_s [kpc]	83.80	93.35	207.73	95.20	100.10
$\rho_{b,0}$ [10^5 M$_\odot$ pc^{-3}]	0.121	0.533	0.906	0.1200	0.1250
r_b [kpc]	0.03	0.03	0.03	0.03	0.03
Σ_0 [M$_\odot$ pc^{-2}]	160.56	213.65	79.81	35.30	35.9
R_d [kpc]	4.41	4.92	2.94	3.14	2.014

TABLE V : Fitted 6-D Parameters after introducing Bulge density) for comparison of VM-DM parameter see FIG.4

Introducing Bulge Density and Kinematics with RC:

All the analysis before this section was done without introducing bulge density and corresponding correction in our model. We performed Monte Carlo Markov Chain using Bhattacharjee datasets for 3 different LSRs but without any prior (flat prior) but introducing bulge density, again by doing that we have seen a significant shift in 6 parameters tabulated in Table V. Further, within the same model using Bhattacharjee 8.0 data, if we combine Eilers data, the combined fitted rotation curve is shown in Fig 3. The mean VM parameter reduced drastically, but again, within the constraint of a 10 percent error bar of the combined Eilers+Bhatt dataset, we recover the similar (Rho DM, Sigma) parameters.The MCMC of 6-D parameter space is shown in Fig. 3 Similarly, we have also combined Mroz rotation curve for cepheids with Bhattacharjee(8.0,200),noting that bulge information is only coming from bhattacharjee's dataset . The circular velocity of a test particle in the Galaxy is, of course, not a directly measured quantity. The RC of the Galaxy has to be derived from the kinematical as well as positional data for an appropriate set of tracer objects moving in the gravitational field of the Galaxy. Except in few cases, the full three-dimensional velocity information of the tracers is not available, and the RC has to be reconstructed from only the measured line-of-sight (LOS) velocity and positional information of various tracer objects in the Galaxy.Since currently not much reliable observational information on β is available, in this article we calculate the circular velocities using the Jeans equation with the velocity anisotropy β of the tracers taken as (1) a radially constant free parameter varying over a possible range of values from $\beta = 0$ (corresponding to complete isotropy of the tracers' orbits) to $\beta = 1$ (corresponding to completely radial orbits of the tracers), (2) a radially varying β of the Osipkov–Merritt (OM) form (see Binney & Tremaine 2008, pp. 297–298) given by

$$\beta = \left(1 + \frac{r_a^2}{r^2}\right)^{-1},$$

where r_a is the "anisotropy radius," and (3) a radial profile of β obtained from recent large high-resolution hy-

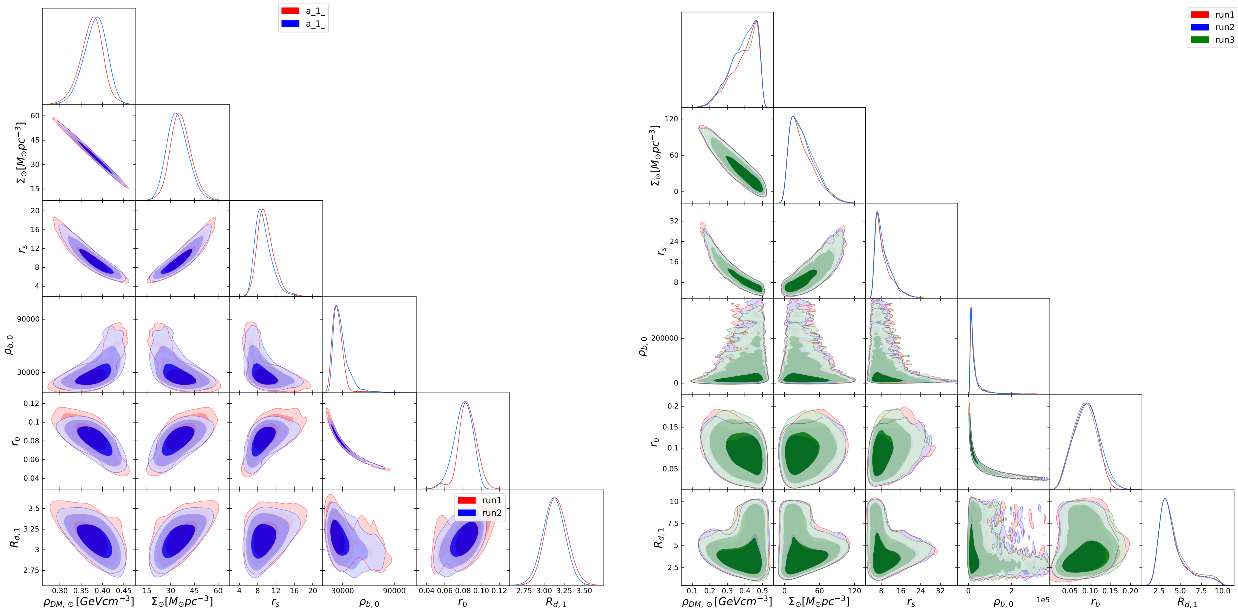

FIG. 3. Posterior probability contours of combined Eilers+Bhatt(β13 dataset(left) and with additional 10% constraint on errorbar(right)

drodynamical simulations of the formation of late-type spirals like our Galaxy (Rashkov et al. 2013) or β13 model . From Fig. 2 and Table IV,V if we combine Bhattacharjee kinematics (>5 kpc) with Eilers TRGB and Mroz Cepheids, both the combined version has degeneracy withing the 2 σ uncertainity with their respective VM-DM parameter space but combining Bhattacharjee's kinematics with >3 kpc Mroz Cepheids shows significant dispacement from its original RC in its parameter space with broken degeneracy as shown in figure below:

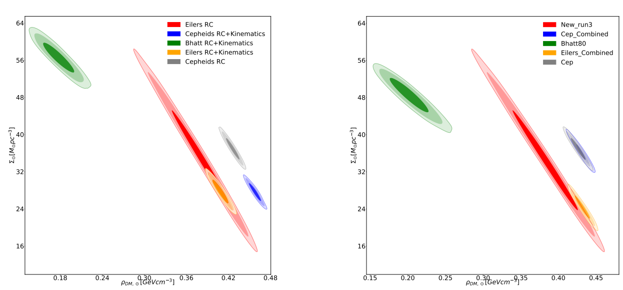

FIG. 4. VM-DM posterior contour comparison after combining different kinematics $z>3$ (left) and $z>5$(right) of Bhatt. et al

Joint Constraint with GAIA DR3 6-D phase plot:

- Finding data from the Gaia archives became the next step to get rotation curves beyond $\sim$ 20kpc. But Gaia parallaxes are reliable ($p/\sigma_p > 10$, i.e. 10% error in the parallax) upto $\sim$5kpc from the Sun. So using Gaia parallaxes alone would only result in a rotation curve upto $\sim$ 13kpc.

- Gaia has detected Cepheids in the Milky Way, in a separate catalogue. The periods can be used to infer parallax from the PL relations. (Metallicity measurements are scarce). The sources can then be cross-matched to the Gaia sources (which has the astrometric and photometric data). RR Lyrae stars may also be used - Gaia parallaxes have been noted to be systematically lower than accurate non-Gaia parallaxes by $\sim$ 0.05 mas (Groenewegen, 2018; Riess et al., 2018). (See also Ripepi et al. (2019), which confirms this under-estimated zero point for parallaxes using Cepheids) .Ripepi et al. (2019) also finds PL and PLZ relations which may be used.Complications with using PL relations includes: Non-fundamental mode oscillators;Possible Binaries; Possible errors in the period determination by the Gaia pipeline (Groenewegen, 2018). If we limit ourselves to the Gaia parallaxes, we can then use the sources with measured radical velocities. We can now construct a rotation curve. Cuts were made on the parallax errors (within 10%), distance from the Galactic Plane ($<$ 0.5kpc) and vertical velocity component $\left(W_s < 50 \text{ km s}^{-1}\right)$

This was also compared with a rotation curve constructed using TRGB stars using Jean's equation for axisymmetric systems (Eilers et al., 2019). Xue et al. (2011) gives LOS velocities and distances to 4985 Blue Horizontal Branch (BHB) stars, of which 4982 were cross-matched with Gaia DR2 (for accurate positions and for proper motions). Similarly, from Xue et al. (2014), we get LOS velocities and distances to 6036 K Giants (all crossmatched) and proper motions from Gaia. Using the Sun's Galactocentric Distance (R_0) as 8.122 kpc (Eilers et al., 2019), we calculate the radial distance from the

Galactic Centre as

$$R = \sqrt{R_0^2 + r_h^2 - 2R_0 r_h \cos b \cos l} \tag{31}$$

Note that this R is different from the R in the case of the galactic disk (where it is the projection of the radial vector onto the galactic disk)

We select only the stars which have $R > 20$, and proceed with the calculations. Two binning methods were used, one with a constant number of tracers in each bin (the tracer density then would be determined primarily by the bin width in kpc(dR), and with binning in radial distance (where the tracer density is determined mostly by the number of tracers in each bin). While the first method seems to be useful in the case of lesser number of tracers (as used in Bhattacharjee et al. (2014)), it has the disadvantage that the tracers may not even be in similar radial distances in the same bin (for large number of tracers in each bin (~ 100)).

So, then the choice here becomes the second method (binning by radial distances)

Method 1: Error Propagation in velocities

Errors in velocities are propagated (as with the disk stars) till the final velocity components of V_r, V_θ and V_ϕ The bin sizes were chosen as - BHB Stars: constant 2 kpc bins from 20 kpc to 55 kpc - K Giant Stars: constant 2 kpc bins from 20 kpc to 56 kpc , and then 4 kpc bins out to 68 kpc .

With this, we can estimate the number density n_{tr} and the velocity dispersions σ_R, σ_θ and σ_ϕ, and then anisotropy.The other terms in Eq.13 can be readily obtained by fitting a straight line to log-log plots. The cuts on the data are as follows: - Filter out stars with e_{pmra}=0 (causes a divide by zero error) (This was not considered when we shifted to Monte Carlo).

- Filter out any stars with $R <$ 20kpc (we want to exclude the disk) (automatically done with binning) - Due to the small number of tracers in the last few bins (~ 10), it may be that the standard deviation (i.e. the dispersion of velocities) is over or under-estimated.

It was then realised that the errors in pmra, pmdec, etc. were too high to propagate linearly (the current procedure). So then it was Monte-Carlo to estimate the errors, from the beginning.

Method 2: Monte Carlo for Error Propagation

We assume a normal distribution of measured quantities (ra, dec, dist, pmra, pmdec, rv) with their measured values as the means and errors as the standard deviations. Note that the current implementation assumes no correlation between these quantities. Then we go through the process of calculating σ_R, σ_θ and σ_ϕ. Then we repeat

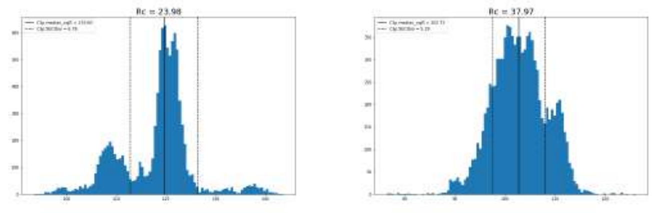

FIG. 5. (a) σ_R for the binning Rc = 23.98kpc is distinctly non Gaussian , (b) σ_R for the binning Rc = 37.97 looks closer to a Gaussian but still not close

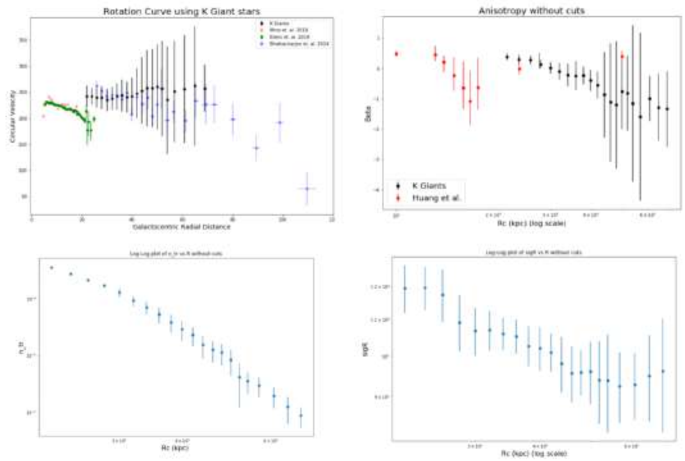

FIG. 6. Rotation Curve($V_c(r)$) and anisotropy parameter($\beta(r)$) construction of Gaia K giants without metallicity cuts, the log-log plots of tracer density vs. r and dispersion vs. r is shown in the next row

this measurement several times to get a distribution of $R, \sigma_R, \sigma_\theta$ and σ_ϕ. And so, we can find the errors on quantities derived quantities.

The propagation of errors will likely not result in a Gaussian distribution for the velocity dispersions, and this is exactly the case. A few sample distributions (not normalized) are shown below:

Looking from the equation Eq.13 due to logarithmic dependency of derivative it is suitable to fit power law with respect to $n_{tr}(r)$ vs. r and $\sigma(r)$ vs.r .As there is a rapid decline of tracer objects within 25 kpc $\pm$ 10 kpc we have only fitted within 25 to 200 kpc to avoid overfitting due to exponentiaal dominance . A major concern earlier on when analysis was still being done with the linear error propagation was that the anisotropy values were consistently lower than expected (from Sarah Bird). The distribution offers a partial explanation. To deal with the outliers and multiple peaks and find a median and standard deviation from them, the `sigma_clipped_stats` from astropy was used for all the quantities. (`sigma=3 and maxiters = 100`) Now, all that is left is finding the slopes and hence, circular velocity (V_c). The results are summarized in the following figures(FIG.6).

Now, we use some cuts (same as Sarah Bird) on the tracers, namely the metallicity [Fe/H] < -1.3 dex, and z > 5kpc. The results are as follows (FIG. 7):

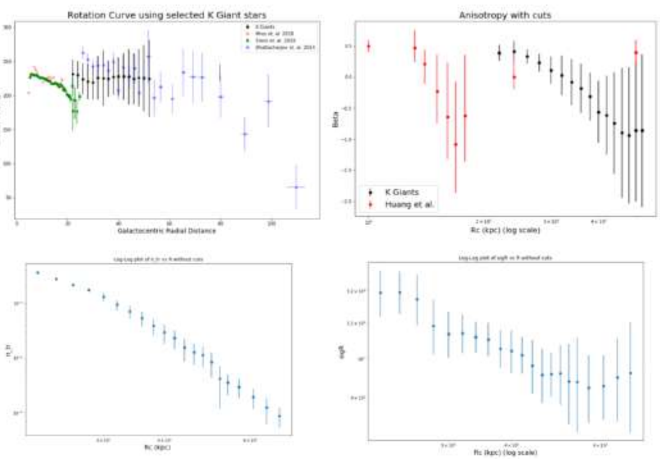

FIG. 7. Rotation Curve($V_c(r)$) and anisotropy parameter($\beta(r)$) construction of Gaia K giants with metallicity cuts, the log-log plots of tracer density vs. r and dispersion vs. r is shown in the next row

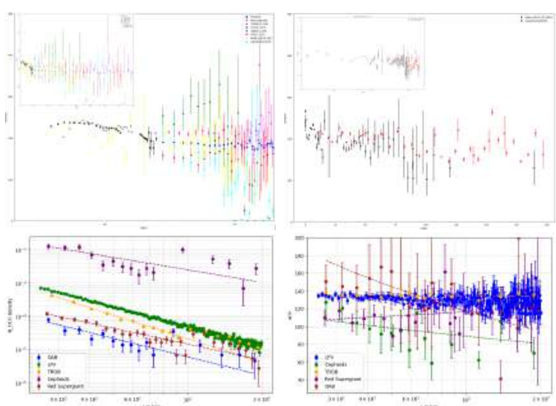

FIG. 8. Rotation Curve($V_c(r)$)constructed for r>25 kpc using Jeans Equation using the same LSR as Eilers($[R_0, V_0] = [81kpc, 228km/s]$ with selected objects from (Gaia archive) compared with similar binning criteria

We first downloaded the entire catalogue of Gaia sources with radial velocity measurements (Gaia Archive). This is a set of 5 files totalling 7.5 GB . Relevant columns for analysis are selected.6D-phase space information and (l,b,radial velocity) to analyze Jeans equation for r > 25kpc .The following cuts are made:
`source_id`, `ref_epoch`
`ra`, `ra_error`, `dec`, `dec_error`
`parallax`, `parallax_error`, `parallax_over_error`
`ra_dec_corr`
`ra_parallax_corr`, `ra_pmra_corr`, `ra_pmdec_corr`
`dec_parallax_corr`, `dec_pmra_corr`, `dec_pmdec_corr`
`parallax_pmra_corr`, `parallax_pmdec_corr`
`pmra_pmdec_corr`
`phot_g_mean_mag`, `bp_rp`, `bp_g`, `g_rp`
`radial_velocity`,
`radial_velocity_error`, `rv_nb_transits`

Concerns regarding the process are 1. Gaia RV has very few sources beyond 25 kpc and $z = 1$kpc 2. Calculation of dispersions are straightforward. However, estimating the errors on this value may not be, since our astrometric measurements may be correlated. 3. Given the lack of data at higher z, it may be required to crossmatch survey data with photometric or spectroscopic distances and radial velocity measurements. 4.White Dwarfs and Globular Clusters are excellent candidate for probing rotation curve and dark matter density but Gaia DR3 doesn't contain radial velocity information and no photometry information available for White Dwarfs exclusively needed for our analysis.Also there are only 156 Globulat clusters for which Binning criteria will not be applicable.

Possible Solutions and conventions to adapt is selection of specific objects like TRGBs(Old stars), Variable Cepheids , Globular Clusters ,OAB spectral type stars , Red Supergiants , Long Pulsating Variable (LPV) stars from Gaia DR3 which already has seperate catalogs according to Gaia Archive data pipeline. Looking for SDSSDR13/Segue /APASS/2MASS/ALLWISE/ crossmatched/TYCHO/URAT/GSC all star crossmatched Sources We used Python package pyia to work with Gaia data. Can possibly provide a solution for the error propagation.for usage of `Generator.multivariate_normal()`.Use the official Gaia documentation to process the transformations (link). As we have In total almost 3 Million objects we have used numba to parallelize and optimize codes and Ultra nest as well as Multinest to perform all MCMC simulations. All the fitting parameters and binning criteria for GAIA DR3 crossmatched all stars is listed in Table IX

We then perform **power law fits** ($n_{tr}(r) \propto r^{-\gamma}$) to the radial profile of the tracer number density for each of the three samples separately. The resulting best power law fits are also shown in Fig. 8. The values of the parameters of the best power law fit for each tracer sample are given in Table VI. Within each sample, there is no significant difference in the values of n_{tr} for the three different sets of GCs, as also seen from the values of the power law fit parameters given in the table.

We have to calculate the galactocentric radial velocity dispersion, σ_r, that appears in the Jeans equation (13), for our non-disk samples. To do this, we first transform the observed heliocentric LOS velocity, v_h, of each individual tracer object to $v_{\rm GSR}$, the velocity that would be measured in the Galactic standard of rest (GSR) frame. This is easily done by correcting for the circular motion of the LSR (V_0) and solar peculiar motion with respect to LSR, $(U_\odot, V_\odot, W_\odot)$.

$$v_{\rm GSR} = v_h + U_\odot \cos b \cos l + V_\odot \cos b \sin l + W_\odot \sin b + V_0 \cos b \sin l. \tag{31}$$

For large samples like the BHB and KG stars described above, we calculate the $v_{\rm GSR}$ for all the individual tracers in the same radial bins as used in the estimation of the

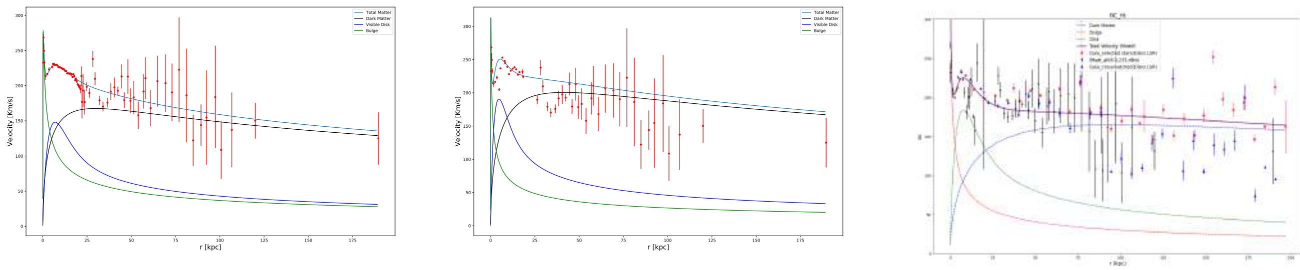

FIG. 9. Fitting of RC combining Bhattacharjee's data with Eilers(leftmost) TRGB ; Mroz's Cepheid(middle) and recent Gaia dr3 selected stars + crossmatched stars (rightmost) .

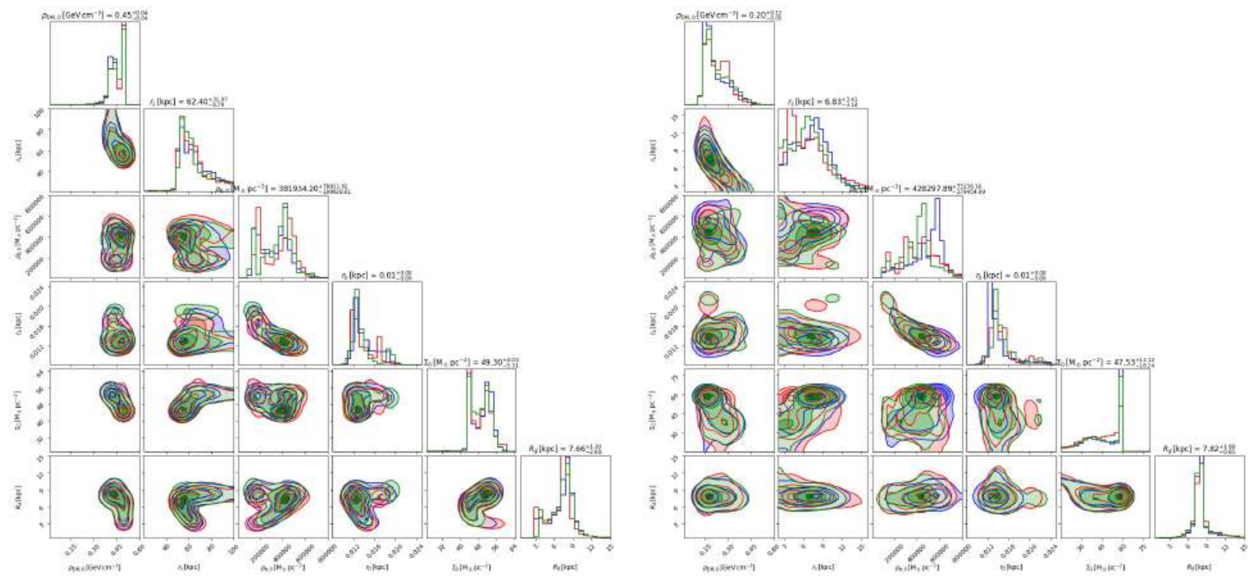

FIG. 10. Posterior probability contours of GAIA DR3 selected objects(left) and GAIA DR3 all stars Crossmatched objects(right)

tracers' number density described above, calculate their dispersion, $\sigma_{\rm GSR}$, and assign it to the mean radius of all the tracers contained within that bin. The corresponding uncertainty, $\Delta\sigma_{\rm GSR}$, in our estimate of $\sigma_{\rm GSR}$ in each bin is calculated by using the standard formula

$$\Delta\sigma_{\rm GSR} = \sqrt{\frac{1}{2(N-1)}}\sigma_{\rm GSR}$$

(Lehmann & Casella 1998; Evans et al. 1993; Graham et al. 1994), where N is the number of objects in the bin.

After we have all equally binned specific objects , again we performed root mean squared binning where

$$v_{\rm mean}^2 = \frac{v_1^2 + v_2^2 + \cdots + v_n^2}{n}$$

and there error as

$$\frac{1}{\sigma^2} = \frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2} + \ldots$$

IV. EXPECTED AND OBSERVED KINEMATICS FOR A SIMPLE MILKY WAY DISK MODEL

We have calculated the expected sky radial velocities and proper motions for stars in an extremely simple model of the kinematics of the Milky Way disk. This is intended to roughly reproduce the motions of stars measured with respect to the solar system barycentre as a function of position on the sky and distance. The application is to fit models of the Milky Way rotation curve to Gaia astrometry and radial velocities.

In this model, the Milky Way is in strict cylindrical differential rotation, with all stars on circular orbits around the Galactic centre, moving at speeds $V_{{\rm c},\star} = V_{\rm c}(R_\star)$, where $V_{\rm c}(R)$ is the circular speed, which depends only on the cylindrical radial coordinate R. The Sun is assumed to move at speed $V_{{\rm c},\odot} = V_{\rm c}(R_\odot)$, the circular speed at the distance from the Sun to the Galactic centre. This model can obviously be approximately correct only in the Galactic plane.

The following schematic diagram shows the steps to

Object	Number of Samples	Binning Criteria	Polynomial Fit Parameters (n_0, γ) $n_{\rm tr} = n_0 \left(\frac{r}{50\,{\rm kpc}}\right)^{-\gamma}$	(σ_0, α) $\sigma_{\rm GSR} = \sigma_0 \left(\frac{r}{50\,{\rm kpc}}\right)^{-\alpha}$
Tip of Red Giant Branch Stars (TRGB)	2,567,955	25 kpc to 95 kpc: 12 bins 95 kpc to 200 kpc: 8 bins	$1.09954 \times 10^{-4}, -2.35047$	$136.436, -0.007$
Globular Clusters	156	No bins	Not applicable	-
Cepheids	2883	25 kpc to 75 kpc: 12 bins 75 kpc to 200 kpc: 6 bins	$58.5 \times 10^{-4}, -1.242$	$104.551, -0.134$
Long Period Variables (LPV)	2,325,775	0 kpc to 200 kpc: 200 bins (uniform binning)	$24.34 \times 10^{-4}, -1.96256$	$133.81, -0.03$
Red Supergiants (RS)	742,263	25 kpc to 100 kpc: 20 bins 100 kpc to 200 kpc: 10 bins	$2.62361 \times 10^{-4}, -1.6051$	$114.37, 0.0604$
OAB Stars	435,848	25 kpc to 75 kpc: 12 bins 75 kpc to 200 kpc: 6 bins	$0.23526 \times 10^{-4}, -1.84126$	$148.592, -0..3044$

TABLE VI :Binnig Criteria and Fitted parameters for GAIADR3 selected stars.

calculate the radial velocity $v_{\rm rad}$ of the star with respect to the Sun, assuming both the Sun and the star are exactly in the plane of the disk. The circular speed is a positive quantity $V_{\rm c}(R) = R|\dot{\phi}|$, where $\dot{\phi}$ is positive for counterclockwise rotation in the right-handed coordinate system.

The Sun and the star are located at $\mathbf{x}_\odot$ and $\mathbf{x}$ and moving with velocities $\mathbf{v}_\odot$ and $\mathbf{v}$. The Sun is located at $(-R_\odot, 0, 0)$. Expressed in terms of the cylindrical coordinates, we have:

$$\mathbf{x}_\odot = \begin{pmatrix} -R_\odot \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} R\cos\phi \\ R\sin\phi \\ 0 \end{pmatrix},$$

$$\mathbf{v}_\odot = \begin{pmatrix} 0 \\ V_{\rm c}(R_\odot) \\ 0 \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} V_{\rm c}(R)\sin\phi \\ -V_{\rm c}(R)\cos\phi \\ 0 \end{pmatrix} \tag{32}$$

The radial velocity of the star with respect to the Sun is given by:

$$v_{\rm rad} = \frac{\mathbf{x} - \mathbf{x}_\odot}{d} \cdot (\mathbf{v} - \mathbf{v}_\odot), \tag{33}$$

which works out to:

$$v_{\rm rad} = \frac{1}{d}\begin{pmatrix} R\cos\phi + R_\odot \\ R\sin\phi \\ 0 \end{pmatrix} \cdot \begin{pmatrix} V_{\rm c}(R)\sin\phi \\ -V_{\rm c}(R)\cos\phi - V_{\rm c}(R_\odot) \\ 0 \end{pmatrix} =$$

$$\frac{1}{d}\left(V_{\rm c}(R)R_\odot - V_{\rm c}(R_\odot)R\right)\sin\phi =$$

$$\frac{R}{d}\left(V_{\rm c}(R)\frac{R_\odot}{R} - V_{\rm c}(R_\odot)\right)\sin\phi. \tag{34}$$

From the drawing, it can be seen that $\sin\phi = (d\sin\ell)/R$ and $\cos\phi = (d\cos\ell - R_\odot)/R$, so that:

$$v_{\rm rad} = \left(V_{\rm c}(R)\frac{R_\odot}{R} - V_{\rm c}(R_\odot)\right)\sin\ell, \tag{35}$$

where:

$$R = \left(d^2 + R_\odot^2 - 2dR_\odot\cos\ell\right)^{1/2}. \tag{36}$$

For the proper motions in ℓ and b, a similar calculation leads to:

$$\mu_{\ell*} = \frac{\cos\ell}{A_{\rm v}d}\left(V_{\rm c}(R)\frac{R_\odot}{R} - V_{\rm c}(R_\odot)\right) - \frac{V_{\rm c}(R)}{A_{\rm v}R}, \quad \mu_b = 0, \tag{37}$$

where $A_{\rm v} = 4.74047...$ km yr s^{-1}, and distance is in kiloparsecs while velocity is in km s^{-1}, resulting in proper motions in mas yr^{-1}.

The above equations hold for all values of ℓ and d. Note the expression for μ_b is zero for stars in the mid-plane of the disk.

For stars off the mid-plane, the following corrections are made: For stars located off the mid-plane (at some height z or, equivalently, latitude b) the expressions should be calculated taken into account that for calculating R the distance in the plane is needed. So if d is still the distance to the star, the values of R, $\sin\phi$, and $\cos\phi$ are to be calculated using $d\cos b$ instead d in the expression for R above. The resulting expression for the radial velocity and proper motions are

$$\begin{aligned} v_{\rm rad} &= \left(V_{\rm c}(R)\frac{R_\odot}{R} - V_{\rm c}(R_\odot)\right)\sin\ell\cos b \\ \mu_{\ell,*} &= \frac{\cos\ell}{A_{\rm v}d}\left(V_{\rm c}(R)\frac{R_\odot}{R} - V_{\rm c}(R_\odot)\right) - \frac{V_{\rm c}(R)}{A_{\rm v}R}\cos b \\ \mu_b &= \frac{1}{A_{\rm v}d}\left(V_{\rm c}(R_\odot) - V_{\rm c}(R)\frac{R_\odot}{R}\right)\sin\ell\sin b \end{aligned} \tag{38}$$

Approximations for $d \ll R_\odot$ and the Oort parameters

Now observations of stars near the sun are considered, where "near" means that $d/R_\odot \ll 1$. From the expression for R above one can write

$$\frac{R_\odot}{R} = \left(1 + \left(\frac{d}{R_\odot}\right)^2 - 2\frac{d}{R_\odot}\cos\ell\right)^{-1/2} \approx 1 + \frac{d}{R_\odot}\cos\ell\,, \tag{39}$$

where higher order terms in $d/R_\odot$ are ignored and the Taylor expansion for $(1-x)^{-1/2}$ was used. The expression for $v_{\rm rad}$ then becomes

$$v_{\rm rad} \approx \left(V_{\rm c}(R)\left(1 + \frac{d}{R_\odot}\cos\ell\right) - V_{\rm c}(R_\odot)\right)\sin\ell\,.$$

We now make a further approximation by expanding to first order the circular velocity curve around its value at $R_\odot$.

$$V_{\rm c}(R) \approx V_{\rm c}(R_\odot) + (R - R_\odot)\frac{\partial V_{\rm c}}{\partial R}\,,$$

where the partial derivative is evaluated at $R_\odot$. Substituting into the expression for $v_{\rm rad}$ leads to

$$\begin{aligned}
v_{\rm rad} &\approx \left(\left(V_{\rm c}(R_\odot) + R\left(1 - \frac{R_\odot}{R}\right)\frac{\partial V_{\rm c}}{\partial R}\right)\left(1 + \frac{d}{R_\odot}\cos\ell\right) - V_{\rm c}(R_\odot)\right)\sin\ell\\
&\approx \left(\left(V_{\rm c}(R_\odot) - R\frac{d}{R_\odot}\cos\ell\frac{\partial V_{\rm c}}{\partial R}\right)\left(1 + \frac{d}{R_\odot}\cos\ell\right) - V_{\rm c}(R_\odot)\right)\sin\ell\\
&\approx \left(\frac{V_{\rm c}(R_\odot)}{R_\odot}d\cos\ell - \frac{R}{R_\odot}\frac{\partial V_{\rm c}}{\partial R}d\cos\ell\right)\sin\ell\\
&\approx \left(\frac{V_{\rm c}(R_\odot)}{R_\odot}d\cos\ell - \frac{\partial V_{\rm c}}{\partial R}d\cos\ell\right)\sin\ell\\
&= \frac{1}{2}\left(\frac{V_{\rm c}(R_\odot)}{R_\odot} - \frac{\partial V_{\rm c}}{\partial R}\right)d\sin(2\ell)\,,
\end{aligned} \tag{39}$$

where in the last step we used that $R/R_\odot \approx 1 - (d/R_\odot)\cos\ell$ and ignored terms with d^2. If we now define the Oort A parameter as:

$$A = \frac{1}{2}\left(\frac{V_{\rm c}(R)}{R} - \frac{\partial V_{\rm c}}{\partial R}\right)_{R=R_\odot}\,,$$

we can write

$$v_{\rm rad} \approx A d\sin(2\ell)\,.$$

Thus for stars near the sun one expects a radial velocity pattern which varies with $\sin(2\ell)$ across the sky and is proportional to the distance to the stars in the sample.

Effect of the sun's peculiar motion

In practice the sun might not follow exactly the velocit pattern dictated by $V_{\rm c}(R)$ but will have a small motion, called "peculiar motion" on top of the velocity resulting from the rotation of the disk. We can write

$$\mathbf{v}_\odot = \begin{pmatrix}0\\V_{\rm c}(R_\odot)\\0\end{pmatrix} + \begin{pmatrix}U_\odot\\V_\odot\\W_\odot\end{pmatrix}\,,$$

where $(U_\odot, V_\odot, W_\odot)$ are the components of the sun's motion along x, y, and z. Substituting this into the first expression for $v_{\rm rad}$ above results in the addition of three components to the radial velocity that only depend on the celestial position of the star. The complete expression for the radial velocity then becomes

$$v_{\rm rad} \approx A d\sin(2\ell) - U_\odot\cos\ell - V_\odot\sin\ell\,,$$

which holds for stars in the mid-plane of the disk, with sun also located in the mid-plane.

Complete expressions for the observed proper motions when $d \ll R_\odot$

Applying the same approximation to the proper motions leads to the introduction of a second Oort parameter B

$$B = \frac{1}{2}\left(-\frac{V_{\rm c}(R)}{R} - \frac{\partial V_{\rm c}}{\partial R}\right)_{R=R_\odot}\,. \tag{1}$$

The complete expressions for the proper motions are

$$\begin{aligned}
\mu_{\ell*} &\approx \frac{1}{A_{\rm v}d}\left((B + A\cos(2\ell))\,d + U_\odot\sin\ell - V_\odot\cos\ell\right)\\
\mu_b &\approx -\frac{W_\odot}{A_{\rm v}d}
\end{aligned}\,.$$

If we consider stars outside the mid-plane of the Milky Way and assumed the still follow the same orbits as stars in the plane (which is reasonable for stars close to the mid-plane of the disk) we can write the expressions for the observed proper motions and radial velocities as:

$$\begin{aligned}
v_{\rm rad} &\approx A d\sin(2\ell)\cos^2 b - U_\odot\cos b\cos\ell - V_\odot\cos b\sin\ell - W_\odot\sin b\\
\mu_{\ell*} &\approx \frac{1}{A_{\rm v}d}\left((B + A\cos(2\ell))\,d\cos b + U_\odot\sin\ell - V_\odot\cos\ell\right)\\
\mu_b &\approx \frac{1}{A_{\rm v}d}\left(-A\sin(2\ell)d\cos b\sin b + (U_\odot\cos\ell + V_\odot\sin\ell)\sin b - W_\odot\cos b\right)
\end{aligned}\,.$$

These expressions are obtained by using $d\cos b$ instead of d as the in-plane distance in the expressions above, and where for the radial velocity an extra projection factor $\cos b$ should be included to account of the projection of the velocity on the line of sight. The factor $1/(A_{\rm v}d)$ for the proper motions remains the same as it is the full distance that determines the magnitude of the proper motion.

In these expressions the terms involving the Oort parameters are approximate while the terms for the sun's peculiar motion are exact (they simply represent the sun's motion projected along the line of sight or along the directions of increasing ℓ and b). [Olling Dehnen (2002)]

General method to calculate the observed radial velocities and proper motions A more general method to obtain

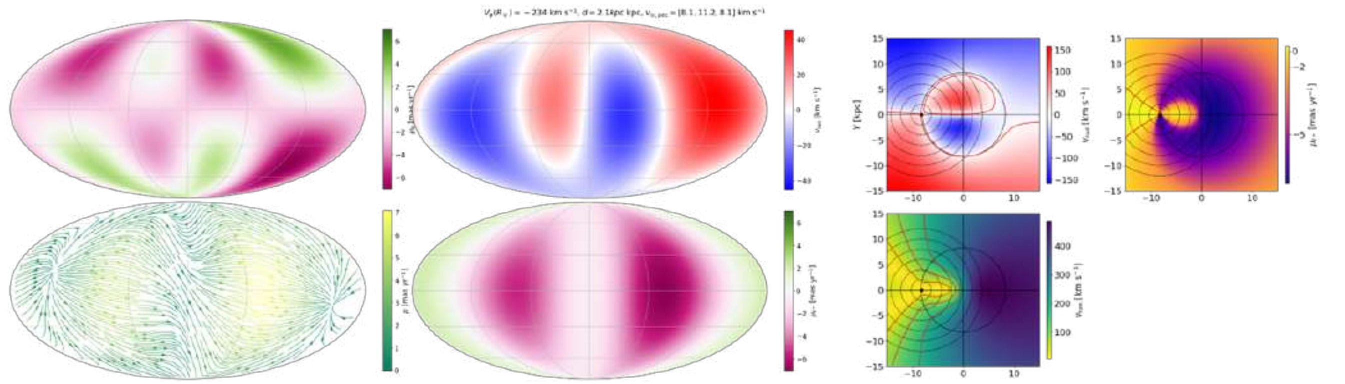

FIG. 11. Proper motion and radial velocity vs sky coordinates (leftmost and middle) The red countours in the radial velocity and proper motion maps are the lines where $v_{\rm rad} = 0$ or $\mu_{\ell *} = 0$. The red countours in the tangential velocity plot indicate the levels 40, 100, 180 km/s.(rightmost)

a complete description of the velocity field in a differentially rotating stellar disk is presented in [Brunetti Pfenniger (2010)]. Here we use vector astrometric methods to achieve the same for the observed velocity field projected on the sky. This method can be applied to any model of the Milky Way and is not restricted to the mid-plane of the disk, or to the sun being located in the mid-plane.

We now consider a star at distance d and Galactic celestial coordinates (ℓ, b). To calculate the position, radial velocity, and proper motions of the star the normal triad $[\mathbf{p}, \mathbf{q}, \mathbf{r}]$ is used:

$$\mathbf{p} = \begin{pmatrix} -\sin\ell \\ \cos\ell \\ 0 \end{pmatrix}, \quad \mathbf{q} = \begin{pmatrix} -\sin b\cos\ell \\ -\sin b\sin\ell \\ \cos b \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} \cos b\cos\ell \\ \cos b\sin\ell \\ \sin b \end{pmatrix} \tag{40}$$

The star is located at:

$$\mathbf{x} = d\mathbf{r} + \mathbf{x}_\odot$$

Its radial velocity and proper motions are then given by:

$$v_{\rm rad} = \mathbf{r}\cdot(\mathbf{v} - \mathbf{v}_\odot), \quad \mu_{\ell *} = \frac{1}{A_{\rm v} d}\mathbf{p}\cdot(\mathbf{v} - \mathbf{v}_\odot),$$

$$\mu_b = \frac{1}{A_{\rm v} d}\mathbf{q}\cdot(\mathbf{v} - \mathbf{v}_\odot), \tag{41}$$

V. MASS MODELLING OF MILKY WAY:

The RC in the disk region is found to depend significantly on the choice of values of the GCs. The RC at large distances beyond the stellar disk, however, depends more significantly on the parameter β than on the values of the GCs. In general, the mean RC is found to steadily decline beyond $r \sim 60\,$kpc, irrespective of the value of β. At any given galactocentric distance r, the circular speed is lower for larger values of β. Considering that the largest allowed value of β is unity (complete radial anisotropy), this allows us to set a model-independent lower limit on the total mass of the Galaxy, giving

$$M(\lesssim 200\,\mathrm{kpc}) \gtrsim (6.8 \pm 4.1) \times 10^{11}\,M_\odot.$$

We have also noted that recent results from high-resolution hydrodynamical simulations of formation of galaxies like Milky Way (Rashkov et al. 2013) indicate an increasingly radially biased velocity ellipsoid of the Galaxy's stellar population at large distances, with stellar orbits tending to be almost purely radial ($\beta \to 1$) beyond $\sim 100\,$kpc. This implies that the above lower limit on the Galaxy's mass (obtained from our results with $\beta = 1$) may in fact be a good estimate of the actual mass of the Galaxy out to $\sim 200\,$kpc.

Given the virial radius and the overdensity convention, the virial mass $M_{\rm vir}$ can be found through the relation

$$M_{\rm vir} = \frac{4}{3}\pi r_{\rm vir}^3 \rho\,(< r_{\rm vir}) = \frac{4}{3}\pi r_{\rm vir}^3 \Delta_c \rho_c.$$

If the convention that $\Delta_c = 200$ is used, then this becomes

$$M_{200} = \frac{4}{3}\pi r_{200}^3 200\rho_c = \frac{100 r_{200}^3 H^2(t)}{G}$$

where $H(t)$ is the Hubble parameter as described above, and G is the gravitational constant. This defines the virial mass of an astrophysical system.

Given M_{200} and r_{200}, properties of dark matter halos can be defined, including circular velocity, the density profile, and total mass. M_{200} and r_{200} are directly related to the Navarro-Frenk-White (NFW) profile given by :

$$\rho(r) = \frac{\delta_c \rho_c}{r/r_s\,(1 + r/r_s)^2}$$

where ρ_c is the critical density, and the overdensity $\delta_c = \frac{200}{3}\frac{c_{200}^3}{\ln(1+c_{200})-\frac{c_{200}}{1+c_{200}}}$ (not to be confused with Δ_c) and the scale radius r_s are unique to each halo, and the concentration parameter is given by $c_{200} = \frac{r_{200}}{r_s}$ In place of $\delta_c\rho_c, \rho_s$ is often used, where ρ_s is a parameter unique to each halo. The total mass of the dark matter halo can then be computed by integrating over the volume of the density out to the virial radius r_{200} :

$$M = \int_0^{r_{200}} 4\pi r^2 \rho(r) dr = 4\pi\rho_s r_s^3 \left[\ln\left(\frac{r_{200}+r_s}{r_s}\right) - \frac{r_{200}}{r_{200}+r_s}\right]$$

$=4\ \pi\rho_s r_s^3 \left[\ln\left(1+c_{200}\right) - \frac{c_{200}}{1+c_{200}}\right]$

From the definition of the circular velocity, $V_c(r) = \sqrt{\frac{GM(r)}{r}}$, we can find the circular velocity at the virial radius r_{200} :

$$V_{200} = \sqrt{\frac{GM_{200}}{r_{200}}}.$$

Then the circular velocity for the dark matter halo is given by

$$V_c^2(r) = V_{200}^2 \frac{1}{x}\frac{\ln(1+cx)-(cx)/(1+cx)}{\ln(1+c)-c/(1+c)}$$

where $x = r/r_{200}$. Although the NFW profile is commonly used, other profiles like the Einasto profile and profiles that take into account the adiabatic contraction of the dark matter due to the baryonic content are also used to characterize dark matter halos. The MW has several baryonic components, including a central nucleus, a bulge, and a disk. While many of their properties remain topics of debate, their masses are reasonably well determined (Bland-Hawthorn and Gerhard 2016). From kinematical and dynamical studies, we know that the mass of the MW must be larger than the sum of the baryonic components; indeed, to maintain the system in stable equilibrium with the observed amplitude of the circular velocity, a large fraction of the MW mass must be invisible, i.e., we cannot measure it directly, but we can infer its presence by its gravitational influence. Despite decades of intense efforts, the estimates of the mass of the MW still show significant scatter. These estimates are very sensitive to assumptions made in the modeling and, in particular, to the shape of the halo in which the Galaxy is embedded. Most mass estimators are limited to the region explored by the available tracer population, whose spatial distribution and kinematics are used to estimate the enclosed mass.

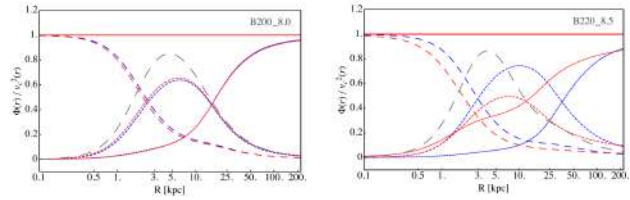

FIG. 12. Comparison of $\frac{\Phi}{V_c^2}$ for Different LSR of Bhatt data .

In the FIG 12 the ratio $\frac{\Phi}{V_c^2}$, where $\Phi(r)$ is the potential of the disc and V_c is the circular velocity (rotation speed),where if $\Phi(r)$ has contributions from other components (like the halo or bulge), this ratio can also help in understanding the relative dominance of the disk potential compared to other components influencing the rotation.The dimensionless ratio $\frac{\Phi}{V_c^2}$ can then be expressed as:

$$\frac{\Phi(r)}{V_c^2(r)} = \frac{\Phi(r)}{r\frac{\mathrm{d}\Phi(r)}{\mathrm{d}r}}.$$

.

Estimates of the MW's mass have been obtained based on the kinematics of halo stars, the kinematics of satellite galaxies and globular clusters, the evaluation of the local escape velocity, and the modeling of satellite galaxy tidal streams. Estimates typically range from as low as $0.5\times10^{12}M_\odot$ to as high as $4\times10^{12}M_\odot$ (Bland-Hawthorn and Gerhard 2016). These estimates assume that dark matter (DM) is in a quasi-spherical virialized halo around the Galaxy (Navarro et al. 1997; Sanders 2010).

We have used the standard NFW profile for dark matter:

$$\rho_{\rm DM}(r) = \frac{\rho_s}{\frac{r}{r_s}\left(1+\frac{r}{r_s}\right)^2},$$

where ρ_s and r_s are the characteristic density and scale radius, respectively. Additional components (e.g., bulge or disk) can be included as needed.

the critical density is given by

$$\rho_{\rm crit} = \frac{3H^2}{8\pi G},$$

where H is the Hubble constant. Define the virial radius r_Δ as the radius where the mean density within a sphere is $\Delta\cdot\rho_{\rm crit}$:

$$\langle\rho(r_\Delta)\rangle = \frac{M(r_\Delta)}{\frac{4}{3}\pi r_\Delta^3} = \Delta\cdot\rho_{\rm crit},$$

where $\Delta = 500$ or 200. we have numerically Computed the mass enclosed within radius r by integrating the total density profile:

$$M(r) = \int_0^r 4\pi r'^2 \rho_{\rm total}(r')\, dr'.$$

$$= \int_0^r 4\pi r'^2 \left[\rho_{\rm DM}(r') + \rho_{\rm bulge}(r') + \rho_{\rm disk}(r')\right] dr'.$$

Iteratively solve for r_{500} and r_{200} by satisfying the mean density condition. The virial mass is:

$$M_\Delta = M(r_\Delta).$$

The concentration parameter is $c_\Delta = \frac{r_\Delta}{r_s}$.

Table VII shows the average values of some of the quantities that physically characterize the galaxy, derived using the mean values of the density parameters for the various datasets.Fig. 4 shows the functional dependence of the mean value of the total mass Mtotal(r) enclosed within a ra dius r for all the datasets and with chioices of at and Gaussian priors.

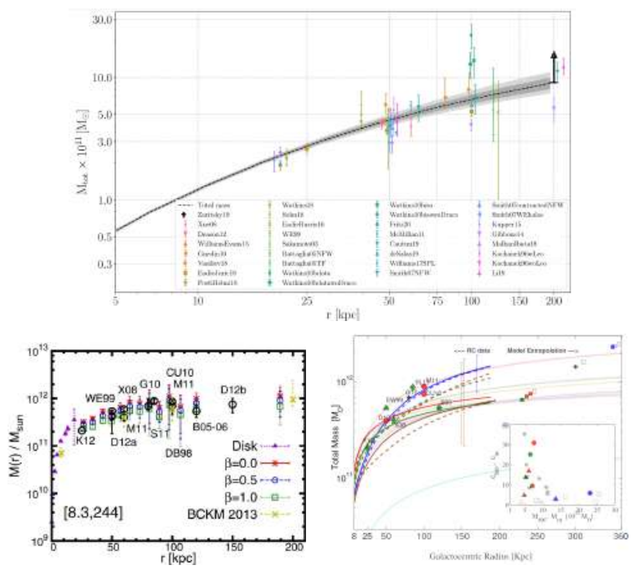

FIG. 13. Comparison of the average value of the Mtotal(r) (halo, disk and bulge) from different references(up) ;for different anisotropy obtained from Bhatt(bottom left) and for all the datasets in our work(bottom right), with at falt (solid) and Gaussian (dashed) priors.Blue Red Green and Brown represents B8.0, B8.3, B8.5 and S8.0 respectively from the Table VII.Triangles represent (M500,c500) pairs and circles represnt (Mvir,cvir) pair.Filled and empty symbols represents Flat and Gaussian priors respectively,where as all gray filled circles represent all datasets from upper Table VII. Included are the masses obtained in various references with symbols - [12], [13], [14], [15], and [16].

In Fig 13 the 1st Row image describes milky Way mass profile for the maximum posterior density parameters (black dashed curve) and the corresponding 68%/95% credible intervals (dark/light gray shade), conditioned on the radius and model-averaged over baryonic morphologies[22]. Also plotted are results from several other studies of the Milky Way's cumulative mass distribution [23-44]. The markers around 50 kpc and 100 kpc are artificially dispersed horizontally so that they are distinguishable. The black arrow denotes the latest lower bound for Mtot from [46]. As with figure 8, note that different choices for R0 and/or V0 among studies can induce apparent discrepancies.

DISCUSSIONS

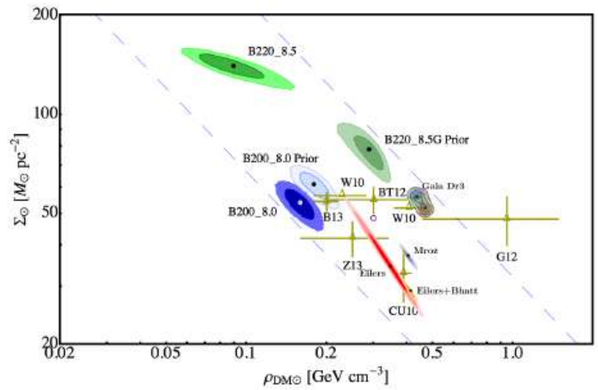

FIG. 14. DM_VM parameter space for different References in comparison with our work.

Fig 4 shows the comparison of some of our results with that of the results obtained in some of other earlier works [12-17] regarding the local DM and VM (disk) densities.This figure underscores the effect of not only the choice of the LSR, the binning and averaging strategies used, and the priors on the visible parameters, but also on the technique used to obtain the best-fit values of the parameters. For instance,Zhang et.al. [14] have analyzed the distribution of the 9000 K-dwarfs in position and velocity space in the vicinity of the Sun, as opposed to the MCMC method used in this work.

It is interesting to note that Catena and Ullio, considering an LSR of 8.33 ± 0.35kpc, 245 ± 10.4 km s^{-1}, and a local disk density prior of $48 \pm 8 M_\odot \mathrm{pc}^{-2}$ have obtained the value of $\rho_{DM\odot} = 0.389 \pm 0.025 \mathrm{GeVcm}^{-3}$, while in our work we have obtained $\rho_{DM\odot} = 0.37 \pm 0.02 \mathrm{GeVcm}^{-3}$ with the Bhattacharjee et.al
dataset corresponding to the LSR of 8.3kpc, 244 km s^{-1}.

The ongoing GAIA mission is working to build a very precise 3D map of our galaxy by surveying over a billion stars, studying their motions and other properties.By combining both horizontal(Coherent) and vertical kinematics this will give us more tighter constraint on to break degeneracy on VM-DM parameters without any prior assumed. This is to significantly reduced the uncertainty in the determination of the LSR, and give a much better estimation of the distribution of visible matter in the galaxy. This, in turn gave tighter constraints on the local DM parameters.

NOTES

For LSR80 - Comparing contour plots of $r_s - \rho_{DM}$, we see that ((i) with external priors, the error bars are thicker for density, since the prior likelihood gives weigtage at point far away from the best fit without prior from data, (ii) as prior goes to smaller disk density, the best fit DM moves to higher values. However, the kinematic data is more affected due to priors. (iii) comparing with fixed zd,

Datasets	$\rho_{DM,\odot}$ [GeV/cm^3]	$\Sigma_\odot$ [M$_\odot$/pc^2]	r_s [kpc]	$R_{D,1}$ [kpc]	M_{200} [M$_\odot$]	c_{200}	M_{500} [M$_\odot$]	c_{500}
Cepheids-β13-nobulge	0.469 ± 0.006	31.1 ± 1.6	16.6 ± 1.1	1.903 ± 0.026	1.34×10^{12}	13.67	1.10×10^{12}	9.43
Cepheids-β13-*Bhatt*	0.426 ± 0.007	37.0 ± 1.5	19.2 ± 1.3	2.156 ± 0.027	1.39×10^{12}	11.97	1.13×10^{12}	8.23
Eiler-nobulge-freevisible	$0.525^{+0.007}_{-0.004}$	$2.78^{+0.98}_{-2.7}$	$3.2^{+0.08}_{-0.17}$		5.36×10^{11}	52.23	4.77×10^{11}	37.02
Eiler-freevisible	$0.091^{+0.031}_{-0.043}$	98.2^{+11}_{-9}	96^{+40}_{-70}	$5.79^{+0.53}_{-0.53}$				
Eiler-run2-*Bhatt*	$0.25^{+0.033}_{-0.013}$	$66.3^{+3.1}_{-8.0}$	76^{+38}_{-56}	$3.478^{+0.006}_{-0.008}$				
Eiler-β13-*Bhatt*-run1	$0.38^{+0.03}_{-0.026}$	35^{+6}_{-7}	$9.5^{+1.4}_{-2.3}$	$3.14^{+0.14}_{-0.14}$	6.39×10^{11}	18.65	5.38×10^{11}	12.98
Eiler-β13-*Bhatt*-run2	$0.332^{+0.014}_{-0.01}$	$48.2^{+3.0}_{-3.4}$	$13.8^{+1.3}_{-1.6}$	$3.194^{+0.061}_{-0.094}$	7.43×10^{11}	13.50	6.09×10^{11}	9.31
Eiler-β13-*Bhatt*-combined	0.432 ± 0.0063	24.3 ± 1.5	6.23 ± 0.51	3.26 ± 0.38	5.64×10^{11}	27.29	4.87×10^{11}	19.14
Cepheids-β13-*Bhatt*-combined-R_0-R_{01}-free	$0.4321^{+0.0061}_{-0.0051}$	$35.9^{+1.3}_{-1.5}$	$18.9^{+1.0}_{-1.2}$	$2.17^{+0.028}_{-0.028}$	1.39×10^{12}	12.16	1.13×10^{12}	8.36
Cepheids-β13-*Bhatt*-combined-R_0-R_{01}-fixed	0.4669 ± 0.0049	26.3 ± 1.1	$13.28^{+0.84}_{-0.95}$	$2.014^{+0.023}_{-0.026}$	1.08×10^{12}	15.88	8.95×10^{11}	11.00
GaiaDR3 − Bhatt − combined (r∼200kpc,with bulge,falt prior:)	$0.45^{0.04}_{-0.04}$	$49.30^{4.03}_{-5.31}$	$62.40^{31.97}_{-8.79}$	$7.66^{1.20}_{-2.60}$	4.22×10^{12}	7.51	3.25×10^{12}	5.07
GaiaDR3 − Bhatt − combined all star crossmatched (r∼200 kpc with bulge)	$0.20^{0.12}_{-0.05}$	$6.83^{3.41}_{-3.14}$	$47.53^{12.12}_{-18.24}$	$7.82^{1.08}_{-0.65}$	4.99×10^{10}	7.71	3.86×10^{10}	5.22

Dataset	M_{500} $(10^{11} M_\odot)$	$M(r_{\max})$ $(10^{11} M_\odot)$	$M_{\rm vir}$ $(10^{11} M_\odot)$	c_{500}	$c_{\rm vir}$
Bhatt. et al. (8.0, 200)	13.30 (13.60)	14.63 (14.79)	25.33 (23.12)	2.54 (2.90)	5.25 (5.91)
Bhatt. et al. (8.3, 244)	9.80 (6.12)	12.46 (6.98)	24.75 (7.55)	0.70 (16.86)	1.62 (30.93)
Bhatt. et al. (8.5, 220)	9.05 (5.17)	11.08 (6.11)	16.22 (6.54)	1.94 (13.62)	4.03 (25.19)
Sofue 2013 (8.0, 200)	3.99 (4.82)	6.60 (6.42)	8.03 (7.07)	1.04 (4.85)	2.24 (9.43)

TABLE VII : Summary of mass and concentration parameters for different datasets.

again affects the kinematic, and pushed kinematic and combined rhoD to higher values
Comparing rhodm-sigma0 plot, it isobvious that DM-VM degenracies dominate kinematic data. Fixing zd, reduces error for kinematic but moves away from RC. By fixing zd or putting priors, kinematic and RC contours move apart, and overlap is only 3 sigma region mostly. With unifirm prior, their 2sigma regions overlap, naturally.Disk with $z > 5$ kpc is better as there is more overlap between kinematic and RC. $z > 5$ kpc disk pushes Kinematic rhoDM to lower values, RC values remain almost same. For Kinematics - thetrajectory from Bovy Tremaine , LSR 80 to LSR 85, $z > 3kpc$ vs $z > 5kpc$ disk, and fixed vs free zd analysis can be done in future.It is worth to note that LSR 85 has larger Error bars.Also in FIG 10 it is worth noting that the halo density and halo radius highly dependent upon the star we are selecting and the number of tracers and the comparatively larger errors in ρ_{DM} occurring due to the consideration of larger radius r $\sim$ 200 kpc unlike others having r $\sim$ 25 to 100 kpc Flat galactic rotation curves can be explained either by introducing a dark matter component, like in the ΛCDM model, or by modifying Newtonian dynamics, like in MOND theories. However, it has been noticed that the acceleration systematically goes down at large distances, i.e. low accelerations. This systematic downward trend in the weak gravity segment of the galactic rotation curves was interpreted as an external field effect within the framework of the MOND theories. However, this feature was observed in the ΛCDM model without invoking any modification of Newtonian gravity, as was first found by one of the authors in the EAGLE simulation .They showed that dark halos in ΛCDM generate an acceleration feature like MOND predicts, but that in the even-larger-radius, even-lower-acceleration regions, the dark halos are saturated and the acceleration then decreases as Kepler's Laws, i.e. Newtonian gravity, but not MOND predict.analyzing Milky ways RC(also M31De-Chang Dai) At very large distances, these data are easier to accommodate in the ΛCDM model.

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Appendix A
ADQL Quesries for fetched Objects

FGKM Spectral Type stars :

```
SELECT vrr.*, gs.pmra, gs.pmdec,
gs.radial_velocity,
gs.parallax, gs.ra, gs.dec, gs.l, gs.b,
gs.parallax_error, gs.radial_velocity_error
FROM gaiadr3.gold_sample_fgkm_stars AS vrr
JOIN gaiadr3.gaia_source AS gs
ON vrr.source_id = gs.source_id
```

OAB Spectral Type stars:

```
SELECT vrr.*, gs.pmra, gs.pmdec,
gs.radial_velocity, gs.parallax, gs.ra,
gs.dec, gs.l, gs.b,
gs.parallax_error, gs.radial_velocity_error
```

```
FROM gaiadr3.gold_sample_oba_stars AS vrr
JOIN gaiadr3.gaia_source AS gs
ON vrr.source_id = gs.source_id
```

TRGB:

```
SELECT source_id, phot_g_mean_mag, bp_rp,
l, b, ra, dec, pmra, pmdec, radial_velocity,
radial_velocity_error, ra_error, dec_error,
parallax, parallax_error
FROM gaiadr3.gaia_source
WHERE phot_g_mean_mag <= 13
AND phot_g_mean_mag >= 8
AND bp_rp >= 1.3
```

Classical Cepheids:

```
SELECT vrr.*, gs.pmra, gs.pmdec,
gs.radial_velocity, gs.parallax, gs.ra,
gs.dec, gs.l, gs.b,
gs.radial_velocity_error
FROM gaiadr3.vari_cepheid AS vrr
JOIN gaiadr3.gaia_source AS gs
ON vrr.source_id = gs.source_id
```

Blue Horizonntal Branch(BHB) stars:

```
SELECT source_id, phot_g_mean_mag, bp_rp,
l, b, ra, dec, pmra, pmdec, radial_velocity,
radial_velocity_error, ra_error, dec_error,
parallax, parallax_error
FROM gaiadr3.gaia_source
WHERE parallax_over_error >= 5
AND parallax > 0
AND (4.74 / parallax * pm) >= 145
AND phot_g_mean_mag
+ 5 + 5 * LOG10(parallax / 1000) < 138.07
* POWER(bp_rp, 6) - 153.85 *
POWER(bp_rp, 5) - 40.727 * POWER(bp_rp, 4)
+ 73.368 * POWER(bp_rp, 3) - 7.4054 *
POWER(bp_rp, 2) - 9.5575 * bp_rp + 3.8459
AND phot_g_mean_mag + 5 +
5 * LOG10(parallax / 1000) > -3.2382 *
POWER(bp_rp, 3) + 7.1259 *
POWER(bp_rp, 2) - 3.583 * bp_rp - 0.2
AND bp_rp < 0.5
AND bp_rp > -0.4;
```

RR Lyrae Stars :

```
SELECT vrr.*, gs.pmra, gs.pmdec,
gs.radial_velocity, gs.ra, gs.dec,
gs.parallax, gs.l, gs.b,
gs.parallax_error, gs.radial_velocity_error
FROM gaiadr3.vari_rrlyrae AS vrr
JOIN gaiadr3.gaia_source AS gs
ON vrr.source_id = gs.source_id;
```

Long Period Variable (LPV) stars or Red Supergiants(RS):

```
SELECT vrr.*, gs.pmra, gs.pmdec,
gs.radial_velocity, gs.parallax, gs.ra,
gs.dec, gs.l, gs.b, gs.radial_velocity_error,
gs.parallax_error, gs.ra_error, gs.dec_error,
gs.pmra_error, gs.pmdec_error
FROM gaiadr3.vari_classifier_result AS vrr
JOIN gaiadr3.gaia_source AS gs
ON vrr.source_id = gs.source_id
WHERE vrr.best_class_name = 'LPV' --or 'RS';
```

Appendix B

Catalog query for Crossmatched sources:

Following tables from the Catalog has been fetched that are already being crosshatched according to Best nearest Neighbor Gaia Pipeline that uses K-Tree Algorithm to crossmatch considering different EPOCH of different sources :

```
gaiadr2.sdssdr9_best_neighbour
gaiadr3.allwise_best_neighbour gaiadr3.apassdr9_best_neighbour
gaiadr3.gsc23_best_neighbour
gaiadr3.panstarrs1_best_neighbour
gaiadr3.ravedr5_best_neighbour
gaiadr3.ravedr6_best_neighbour
gaiadr3.sdssdr13_best_neighbour
gaiadr3.skymapperdr2_best_neighbour
gaiadr3.tmass_psc_xsc_best_neighbour
gaiadr3.urat1_best_neighbour
```

Gaia Selected Stars			**Gaia Crossmatched**		
r (kpc)	v_c	Σ_{v_c}	r (kpc)	v_c	Σ_{v_c}
3.103	200.20	0.129	6.537	232.45	0.001
6.401	234.71	0.002	11.885	223.30	0.003
10.515	215.72	0.001	17.867	211.40	0.640
14.107	212.75	0.151	22.453	193.55	2.565
19.226	201.66	1.042	28.017	189.05	2.588
22.058	192.73	2.808	34.052	193.47	3.241
26.910	206.00	1.484	40.086	178.17	3.707
30.948	214.67	5.071	46.121	202.44	4.429
34.052	191.69	1.991	52.155	192.79	5.513
37.328	211.89	6.420	58.190	193.68	5.812
40.810	195.71	2.247	64.224	180.39	5.988
46.121	207.07	2.672	70.259	160.80	6.992
50.086	213.96	9.114	76.293	221.01	7.572
52.155	198.43	3.017	82.328	179.46	7.263
57.845	199.79	3.199	88.362	181.29	6.463
62.845	212.76	11.790	94.397	105.10	0.045
64.224	210.36	3.592	100.431	122.15	7.087
70.052	181.43	3.805	106.466	102.88	7.317
76.155	218.37	4.059	112.500	110.14	5.379
82.259	181.48	4.535	118.534	142.05	10.738
88.362	186.89	4.804	124.569	153.60	6.172
94.466	159.56	5.246	130.603	224.52	13.065
100.603	169.66	5.344	136.638	106.22	12.929
106.466	184.78	6.068	142.672	144.31	12.071
107.500	209.78	22.625	148.707	105.11	3.510
112.845	156.96	1.665	154.741	139.43	13.745
118.879	145.75	5.640	160.776	134.10	8.724
124.569	170.50	6.723	166.810	144.67	13.357
126.638	220.53	27.217	172.845	200.03	16.294
131.207	151.12	6.225	178.879	73.86	8.220
136.638	175.71	7.856	184.914	110.57	0.836
139.397	160.43	22.826	190.948	95.10	1.706
143.448	151.25	7.029			
148.707	176.70	8.204			
152.155	195.19	27.771			
154.741	263.83	9.002			
160.029	197.65	8.745			
164.914	214.58	39.791			
166.810	166.74	9.461			
172.328	193.40	9.618			
178.578	146.47	4.904			
184.698	162.82	3.169			
190.690	213.40	10.783			
196.810	162.66	33.932			

TABLE VIII :Binned Averaged RC for gaia selected stars and Gaia all stars crossmatched stars.

Object	Number of Samples	Binning Criteria	Polynomial Fit Parameters	
			(n_0, γ) $\mathrm{n_{tr}} = n_0 \left(\frac{r}{50\,\mathrm{kpc}}\right)^{-\gamma}$	(σ_0, α) $\sigma_{\mathrm{GSR}} = \sigma_0 \left(\frac{r}{50\,\mathrm{kpc}}\right)^{-\alpha}$
2MASS	3,000,000	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–75 kpc: 5 bins 75–100 kpc: 4 bins 100–200 kpc: 2 bins each	$8.3018 \times 10^{-5}, -1.6089$	$113.137, 0.046$
AllWISE	3,000,000	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$7.0648 \times 10^{-5}, -1.5722$	$116.124, 0.079$
APASS	3,000,000	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–75 kpc: 5 bins 75–100 kpc: 4 bins 100–200 kpc: 2 bins each	$1.9925 \times 10^{-4}, -0.6533$	$99.637, 0.030$
GSC	3,000,000	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$1.0812 \times 10^{-4}, -1.2691$	$90.764, 0.275$
PanSTARRS	3,000,000	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$1.2640 \times 10^{-4}, -1.2737$	$91.187, 0.279$
RaVE (DR5)	450,659	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$4.1230 \times 10^{-5}, -0.3575$	$110.581, 0.049$
RaVE (DR6)	450,597	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$4.1680 \times 10^{-5}, -0.3294$	$110.277, 0.047$
SkyMapper	3,000,000	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$1.1610 \times 10^{-4}, -0.5780$	$116.882, 0.085$
Tycho	2,518,330	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$8.5110 \times 10^{-6}, -0.1162$	$142.653, 0.070$
URAT	3,000,000	0–25 kpc: 20 bins 25–50 kpc: 10 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins each	$2.7730 \times 10^{-4}, -1.2265$	$104.644, 0.119$
SDSS (DR9) (with Gaia DR2)	3,000,000	0–25 kpc: 15 bins 25–50 kpc: 5 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins	$2.3200 \times 10^{-4}, -1.2289$	$110.114, 0.190$
SDSS (DR13) (with Gaia DR3)	3,000,000	0–25 kpc: 15 bins 25–50 kpc: 5 bins 50–100 kpc: 4 bins 100–200 kpc: 2 bins	$2.4470 \times 10^{-4}, -1.6078$	$92.567, 0.177$

TABLE:IX Summary of catalogs with binning criteria and polynomial fit parameters. The polynomial fit parameters are given as (n_0, γ) and (σ_0, α), following the equations for n_{tr} and σ_{GSR}.

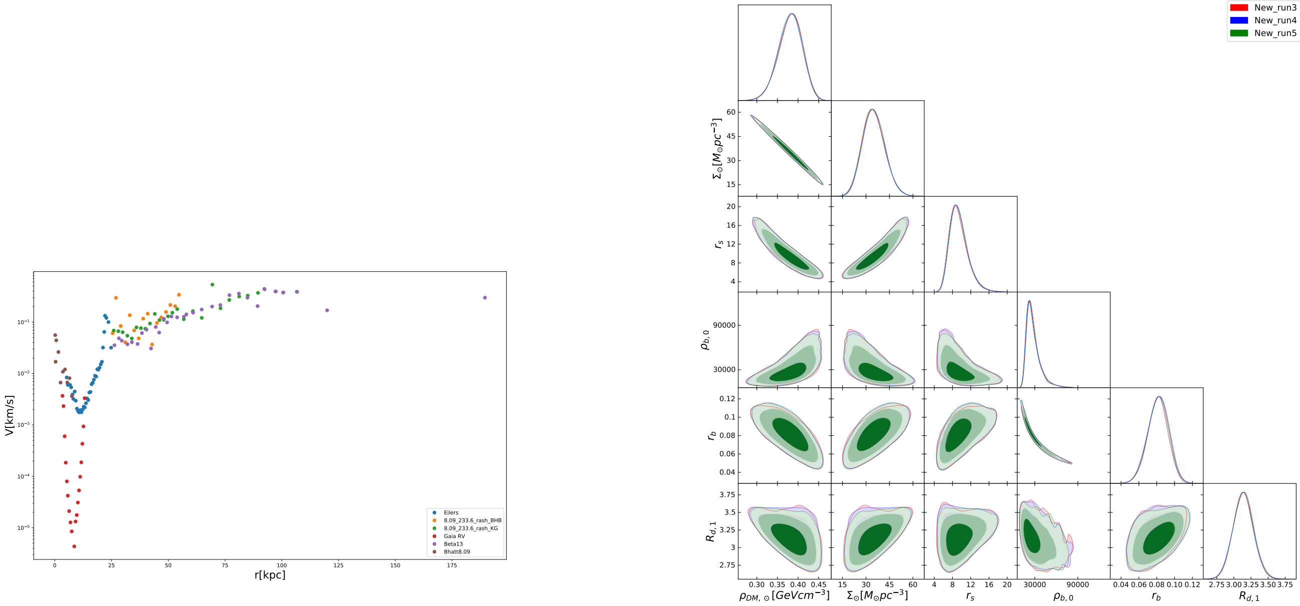

FIG. 15. (Left) RC_all_fractional_error. (Right) triangle_plot_rot_eilers_8.09_233.6_bhatt_beta13_New_run_3_4_5.

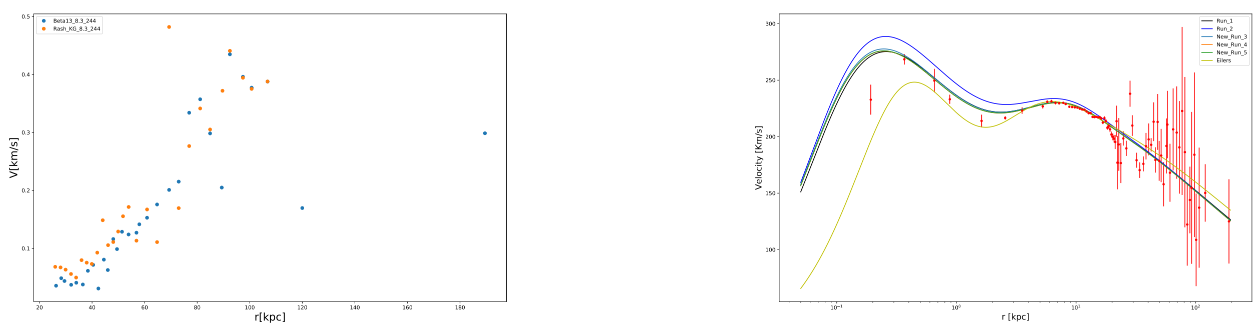

FIG. 16. (Left) RC_beta13_rash_KG_fractional_error. (Right) RC_fit_eilers_bhatt_beta13_8.09_233.6_5_runs_log_scale.

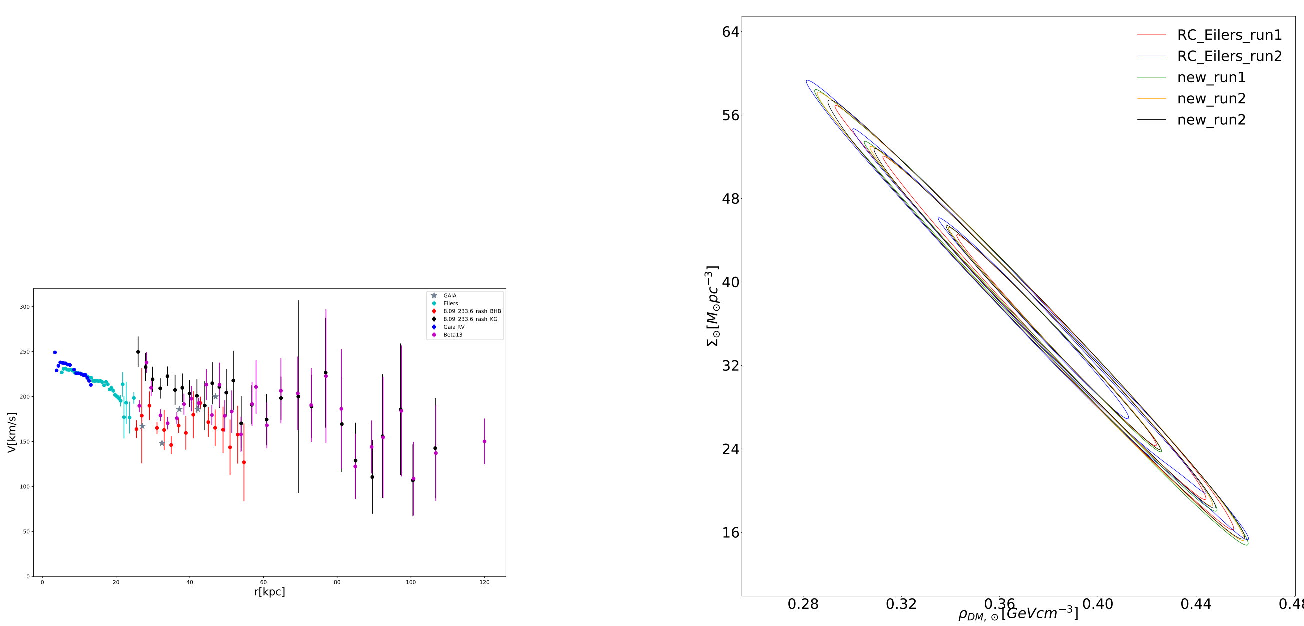

FIG. 17. (Left) RC_all_last_point_removed_beta13. (Right) rho_sigma0_rot_eilers_8.09_233.6_bhatt_beta13_5_runs_version2.

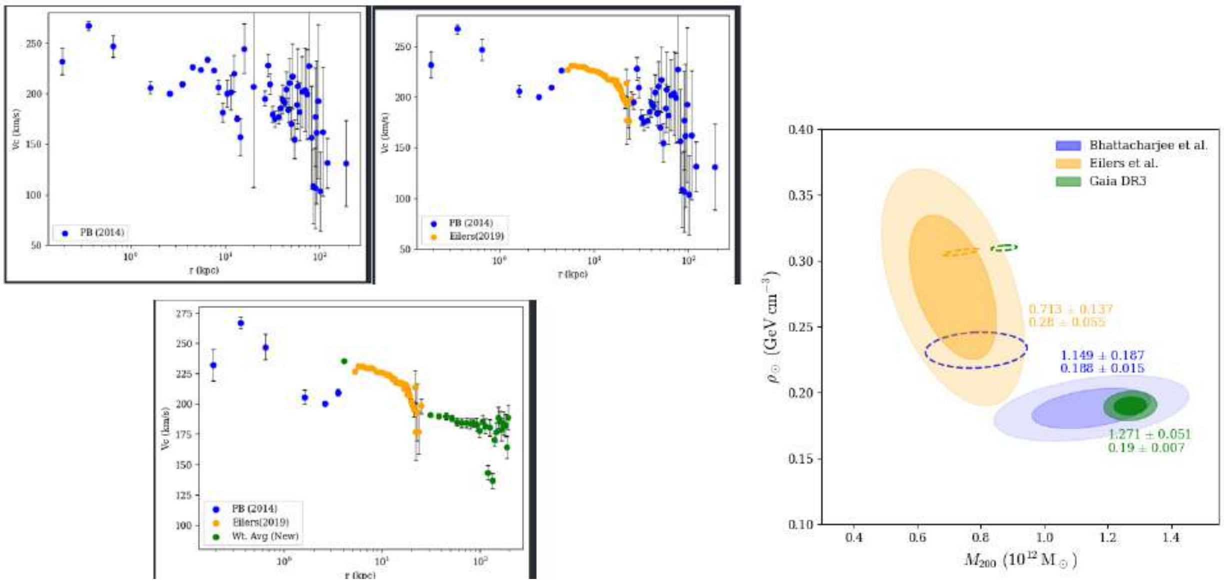

FIG. 20. All RCs and their contour plot with out constraints

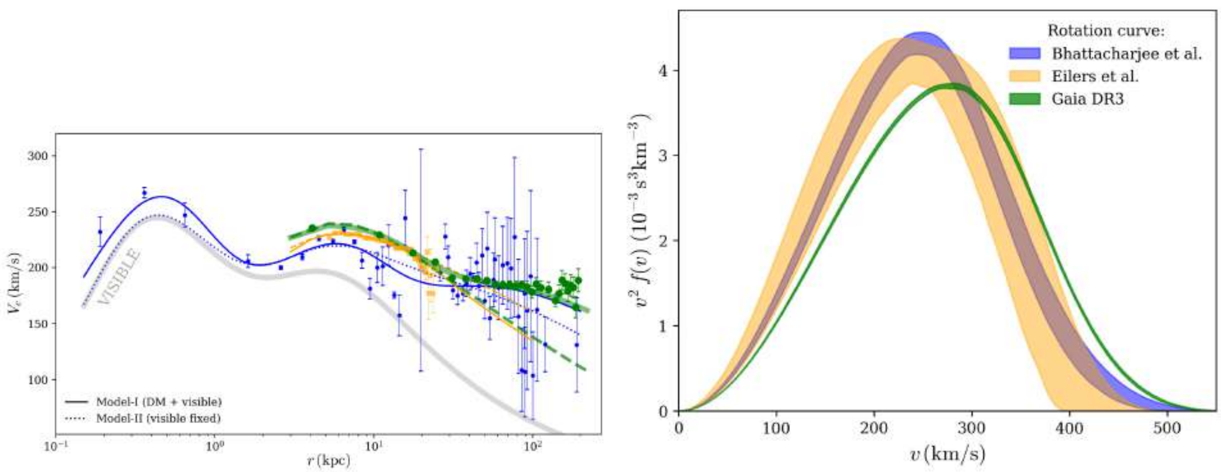

FIG. 21. VDF caompared with standard Halo model(SHM)

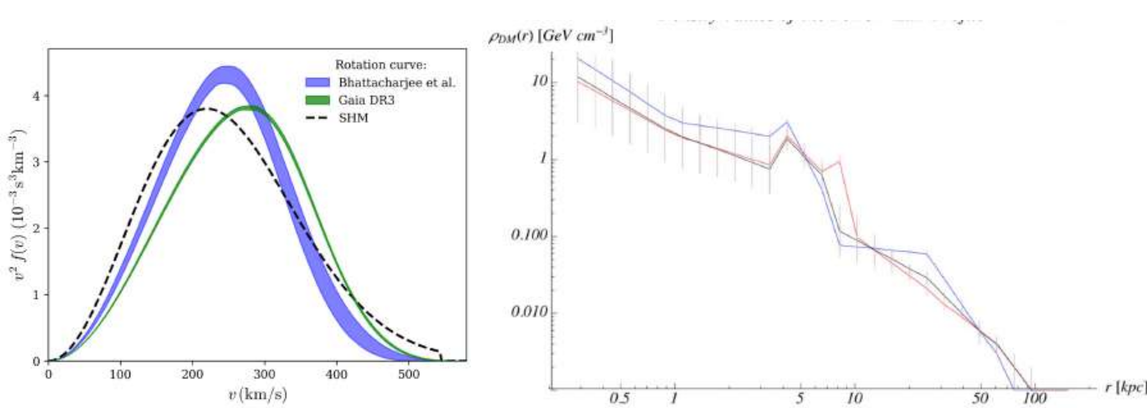

FIG. 22. Left : VDF compared to SHM Right :rho(dm) vs r plot Bhattacharjee for 3 different LSR, why there are two bumps?

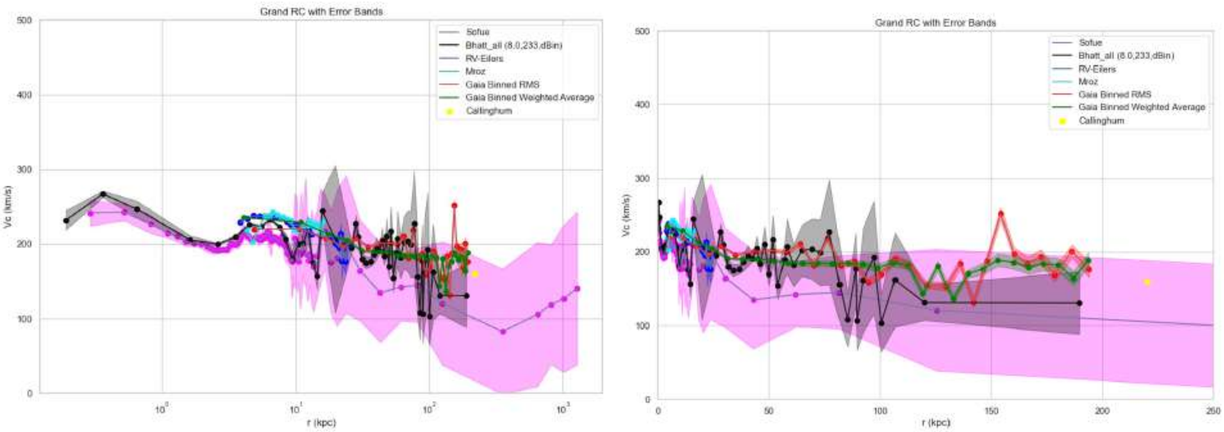

FIG. 23. Grand RC with log scale(left) and Grand RC(right)

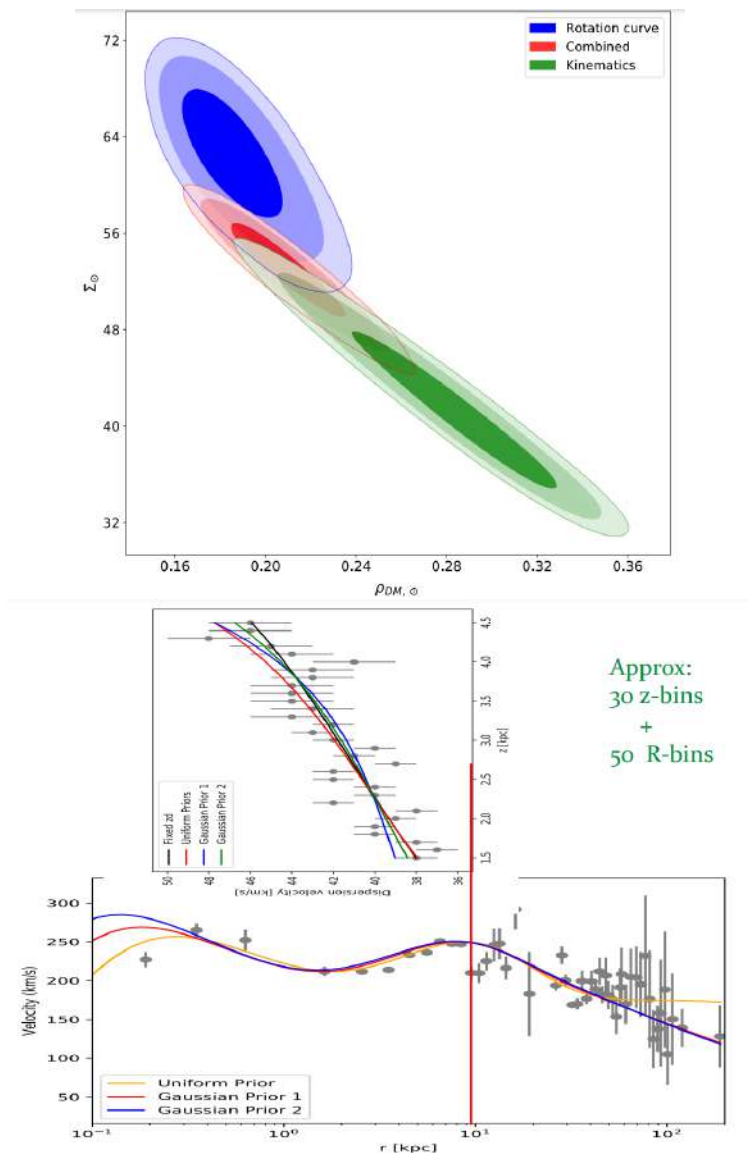

FIG. 24. Kinematics+RC for all datasets(Up) , Vertical and radial joint probe (Down)

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